Jet Quenching in the Smallest Hadronic Collision Systems

Coleridge Faraday

University of Cape Town, South Africa

28^{th} March, Moriond QCD 2026

Based on B. Bert, CF, and W. A. Horowitz, arXiv:2603.21861 [hep-ph] (2026),
CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, arXiv:2512.17832 [hep-ph] (2025),
and **CF** and W. A. Horowitz, JHEP **11**, 019 (2025)

QCD Matter at the Extreme

- High energy collisions of heavy-ions probe the *high temperature* regime of the QCD phase diagram
- QGP formation established in collision of large nuclei

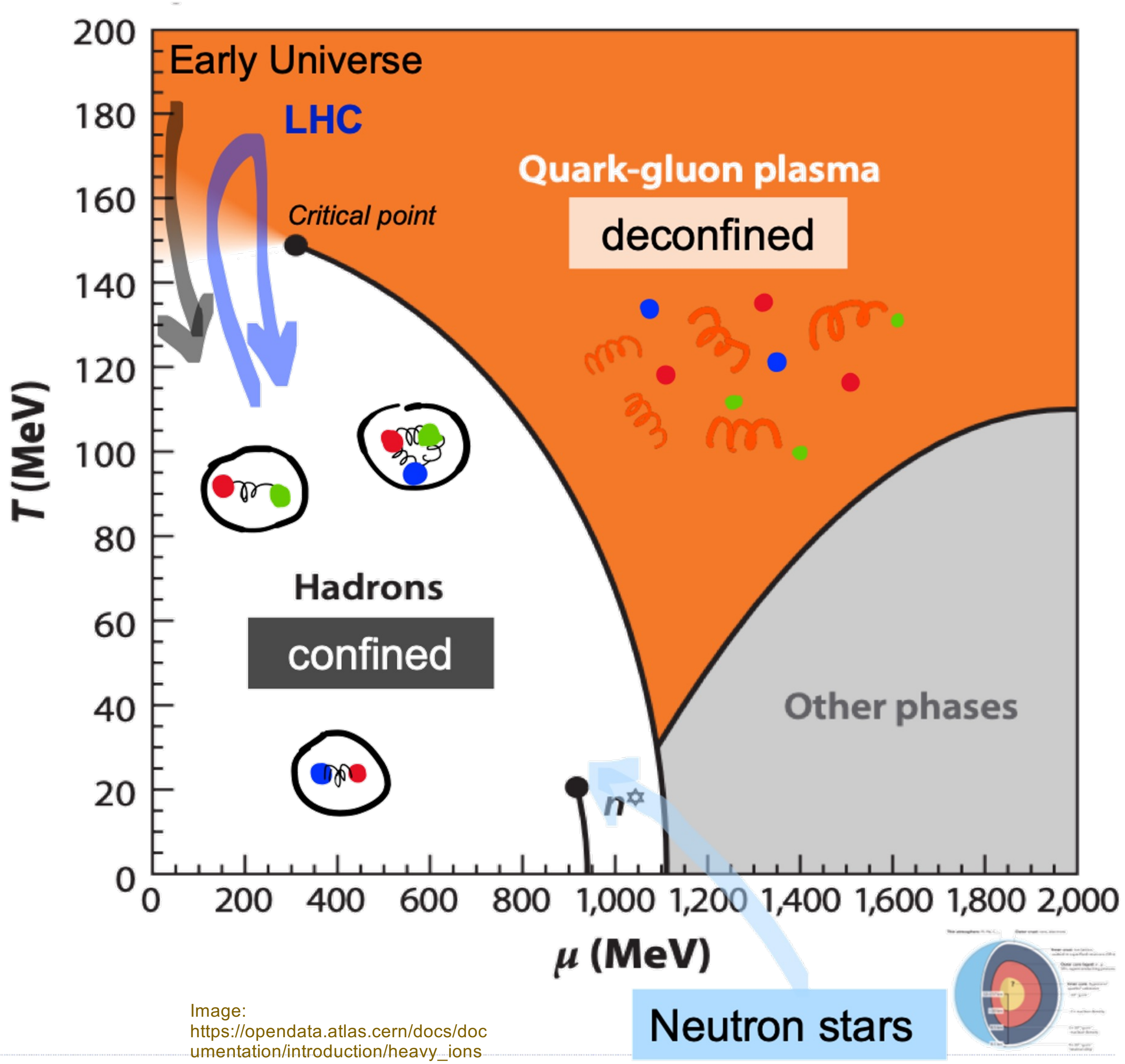

Image: https://opendata.atlas.cern/docs/documentation/introduction/heavy_ions

QCD Matter at the Extreme

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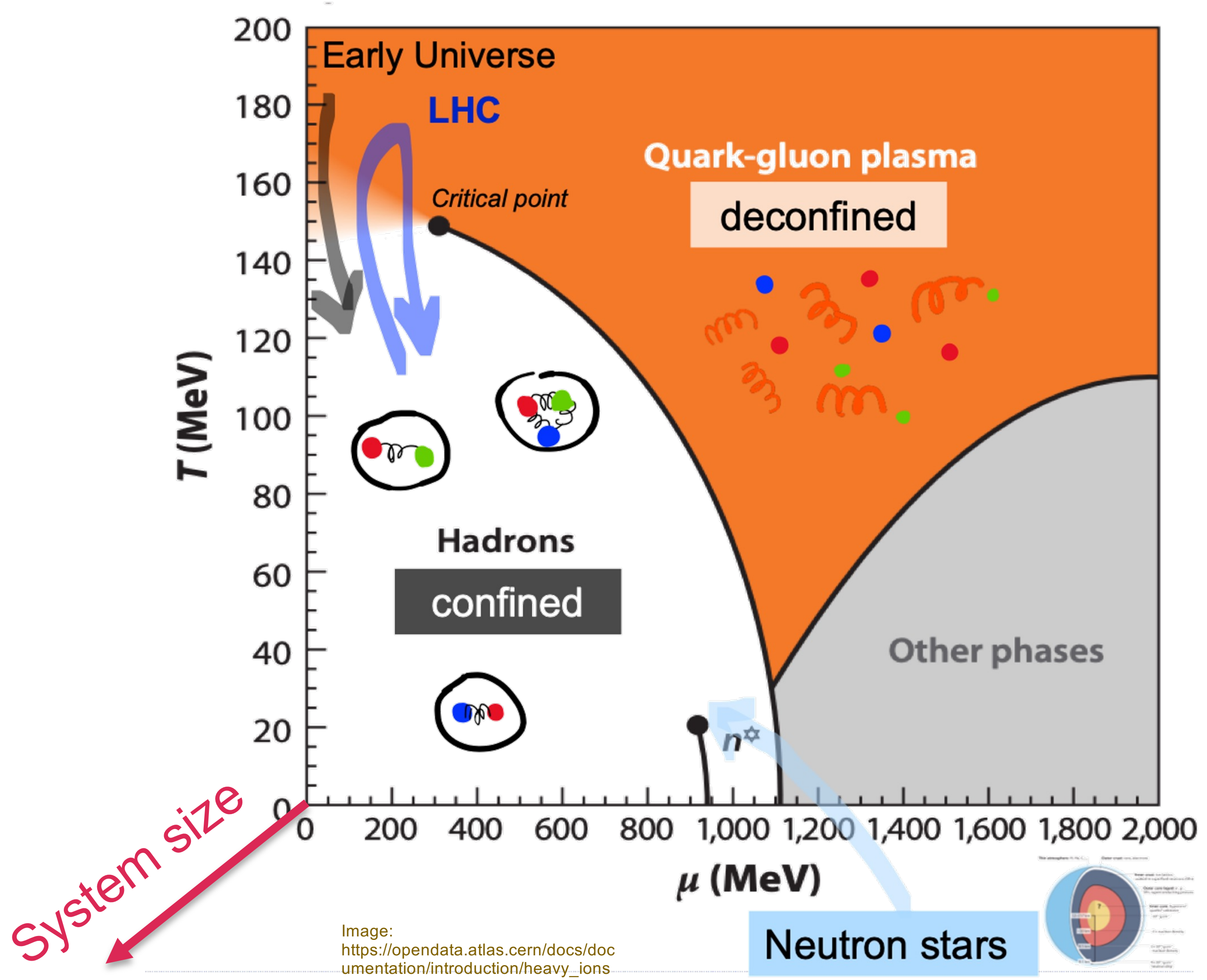
- QGP formation established in collision of large nuclei
- New frontier: **how small can a collision be and still form QGP?**



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Big questions in small systems

- **Does a quark-gluon plasma form in small systems?**

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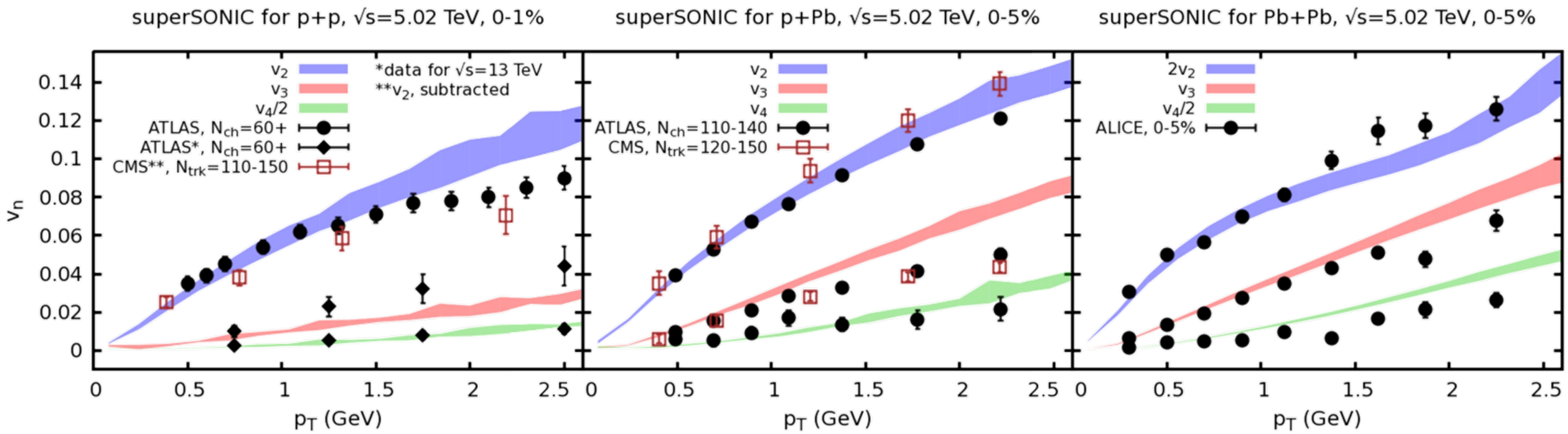
Soft observables say… **YES**
… as long as you're at high enough multiplicity



Weller et al., PLB 774 (2017) 351-356

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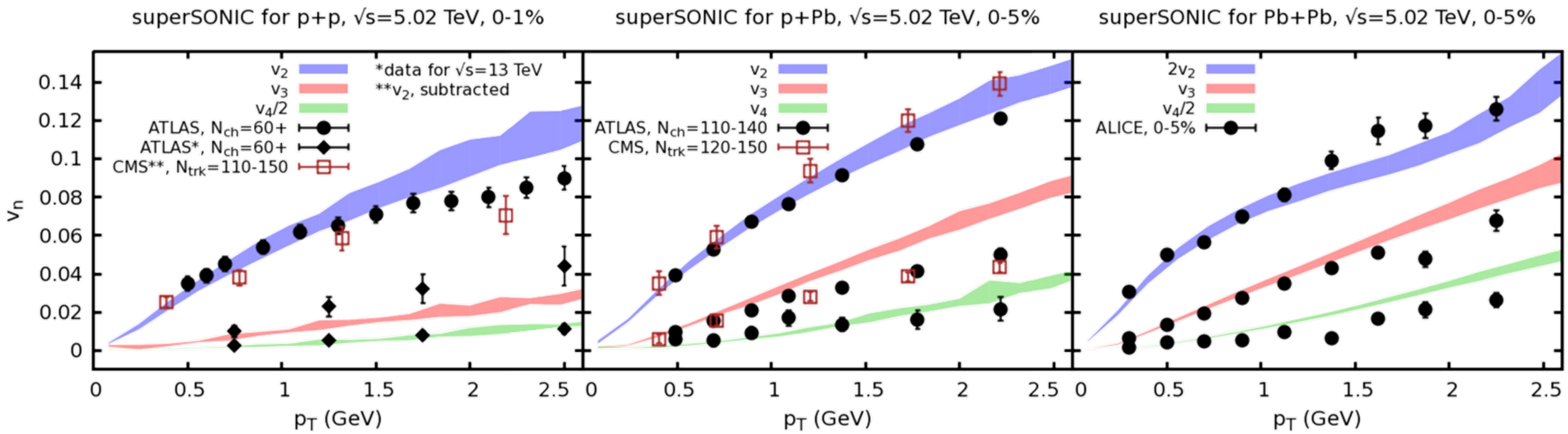
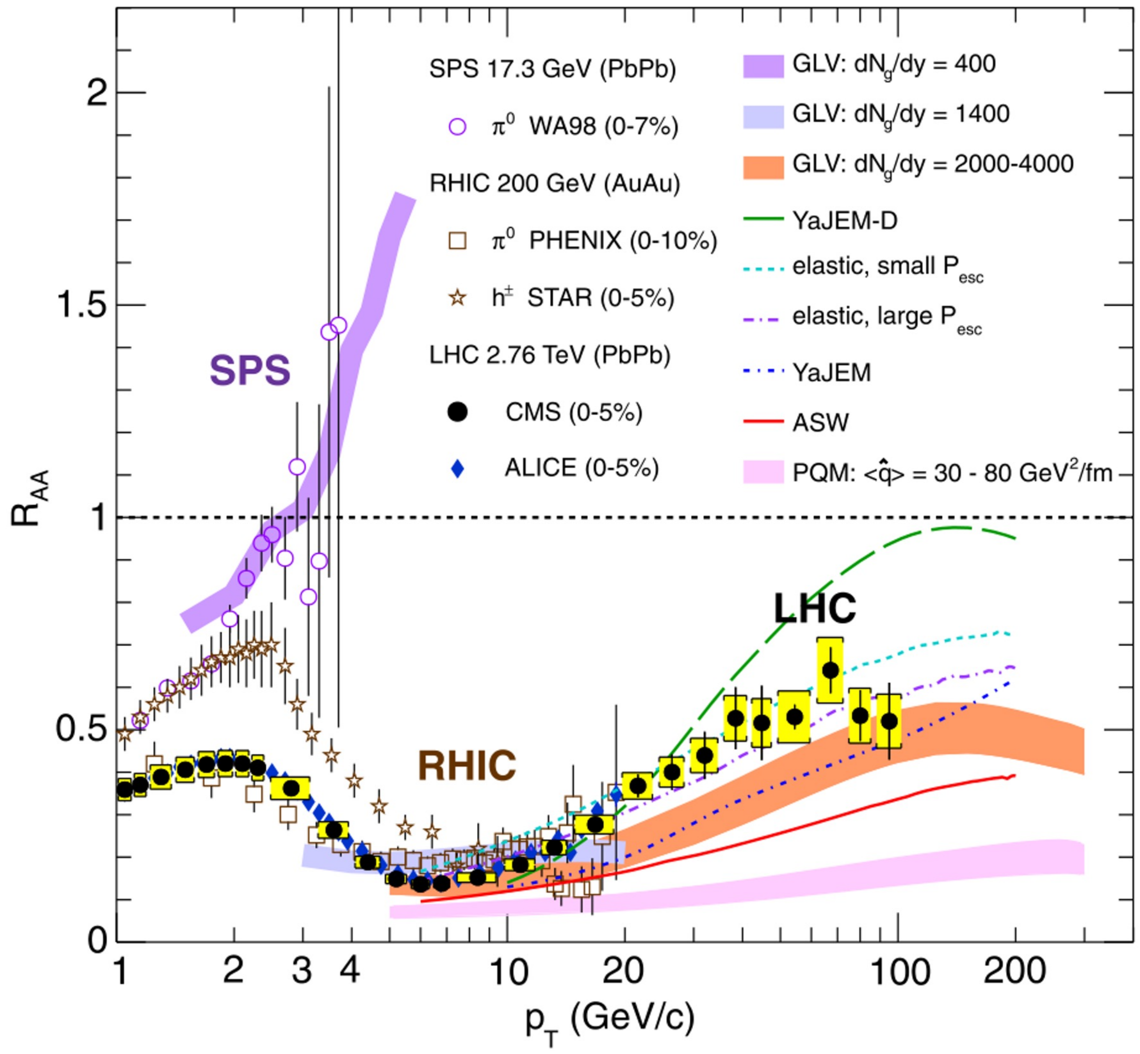


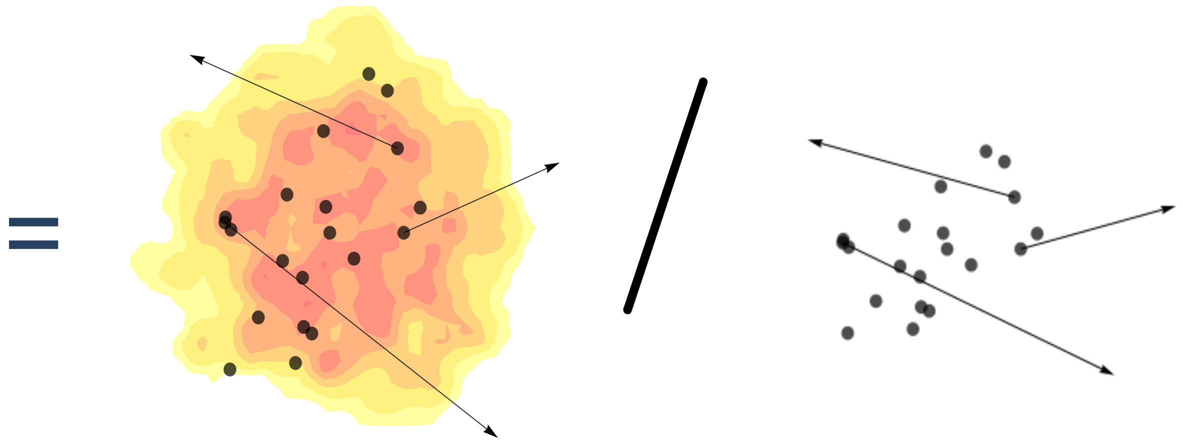
- **How does this QGP differ from that formed in large systems?**

Weller et al., PLB 774 (2017) 351-356

Hard probes of the QGP: R_{AA}



$$R_{AA}(p_T) = \frac{1}{\langle N_{\mathrm{coll}} \rangle} \frac{dN_{AA}/dp_T}{dN_{pp}/dp_T}$$

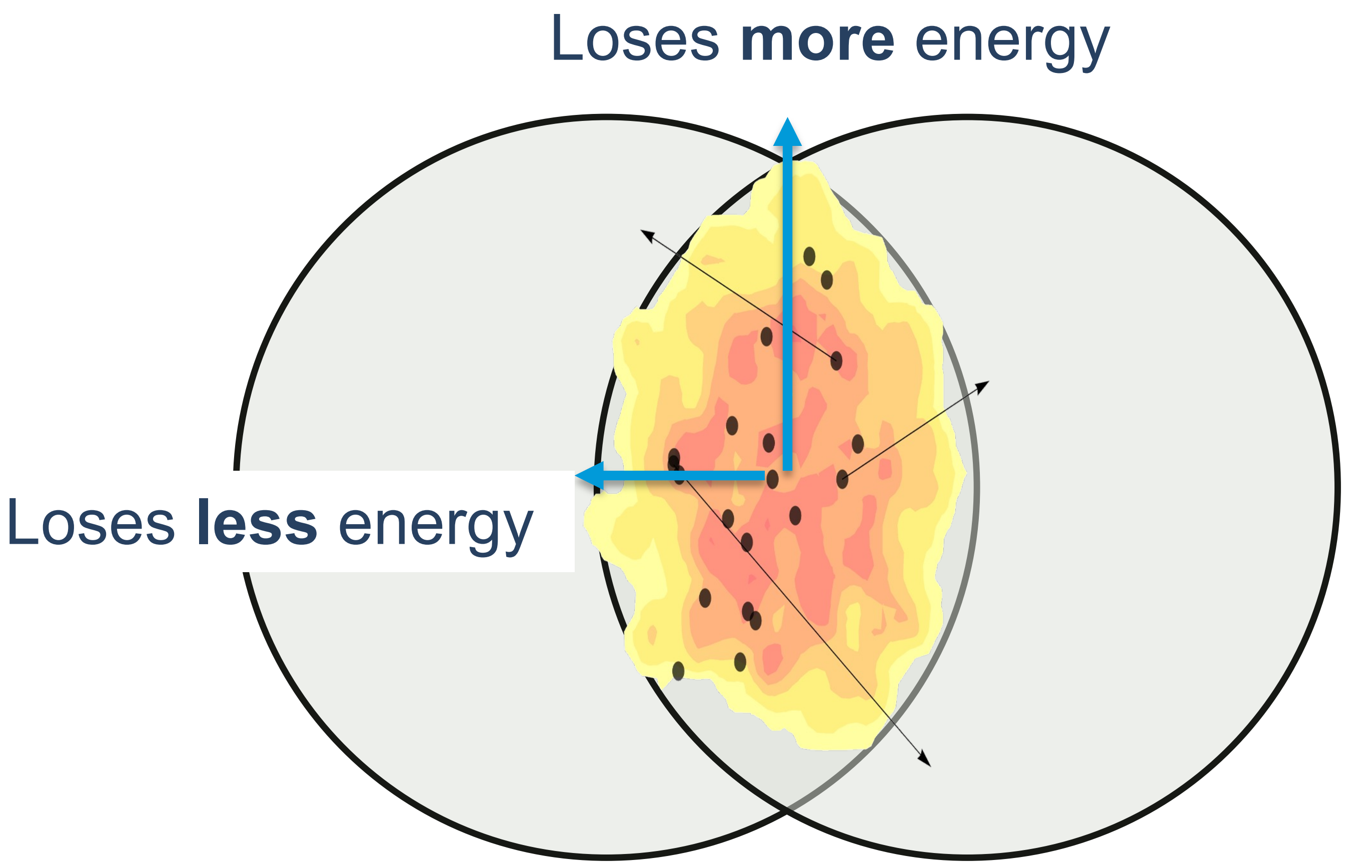


Quantifies how much **energy is lost** by high momentum particles

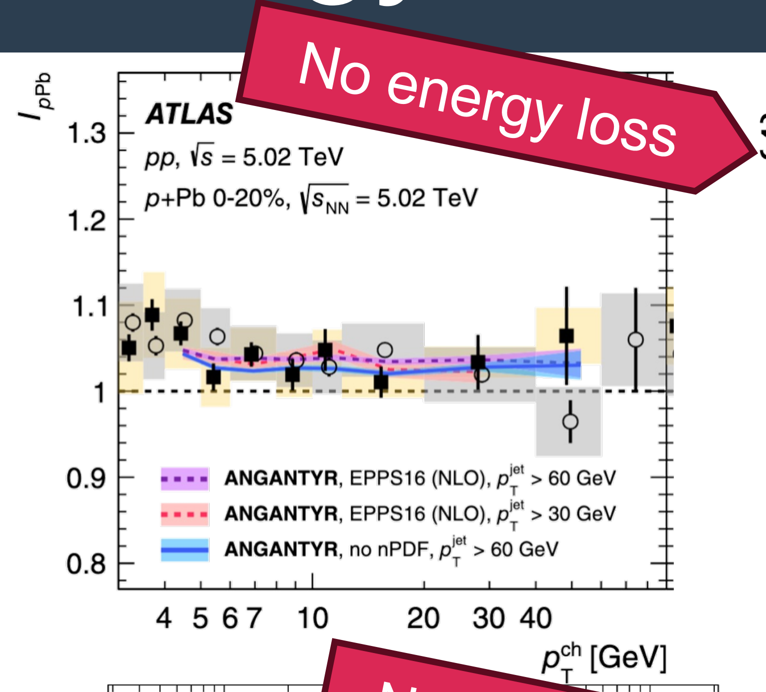
Hard probes of the QGP: high-p_T v_2

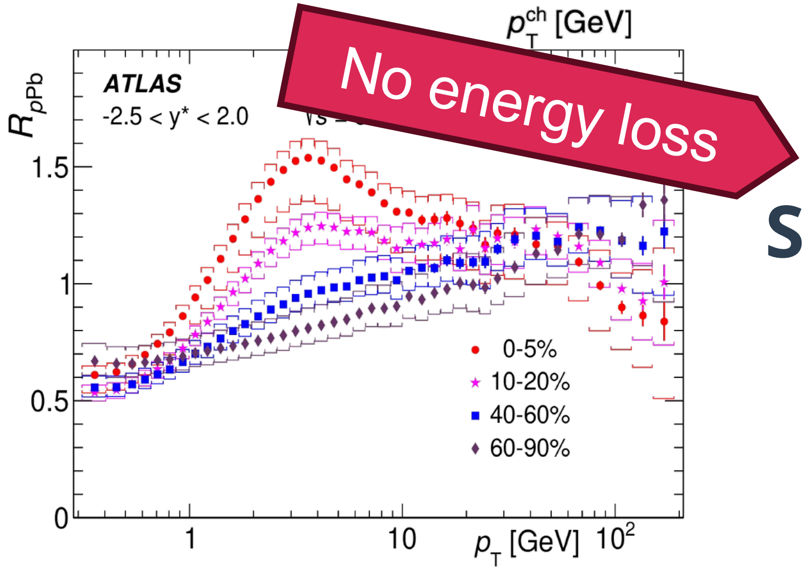
Theoretically, $$\frac{R_{AB}(p_T,\phi)}{R_{AB}(p_T)} = 1 + 2\sum_{n=1}^{\infty} v_n^{\text{hard}}(p_T)\cos\left[n\left(\phi - \psi_n^{\text{hard}}(p_T)\right)\right]$$

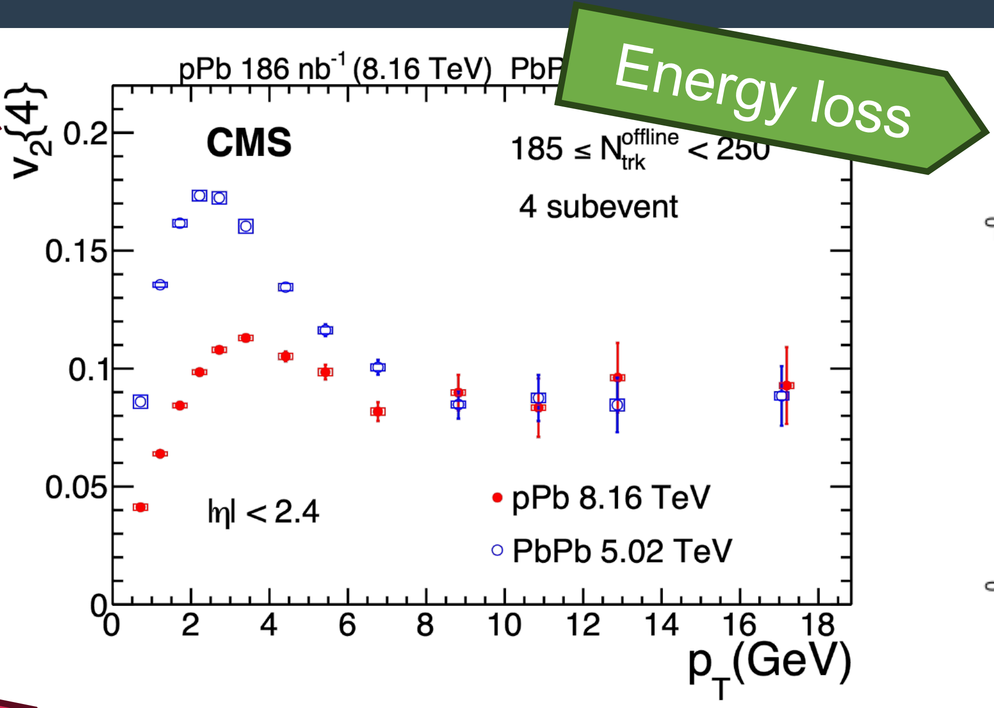
- Captures (in principle) path-length dependent energy loss
- v_2 is dominant (compared to other v_n) in PbPb due to elliptical nature of overlap

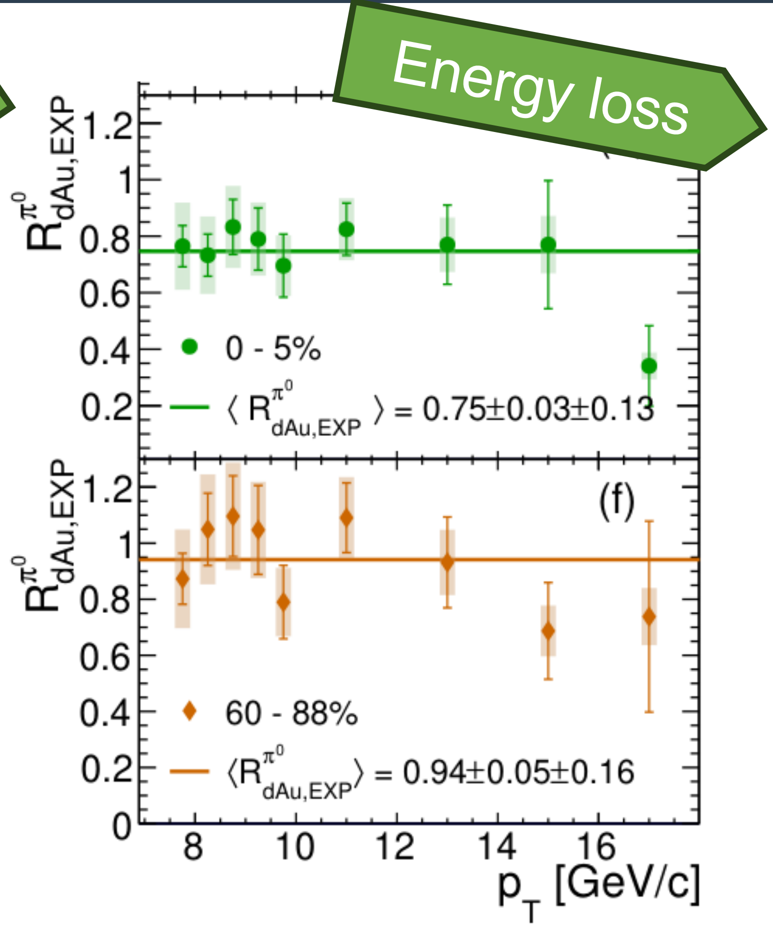


Energy loss (?) in small systems









Small system suppression pattern is not clear

*PHENIX, Phys. Rev. Lett. **134**, 022302 (2025).*
*ATLAS, JHEP 07 (2023) **074***
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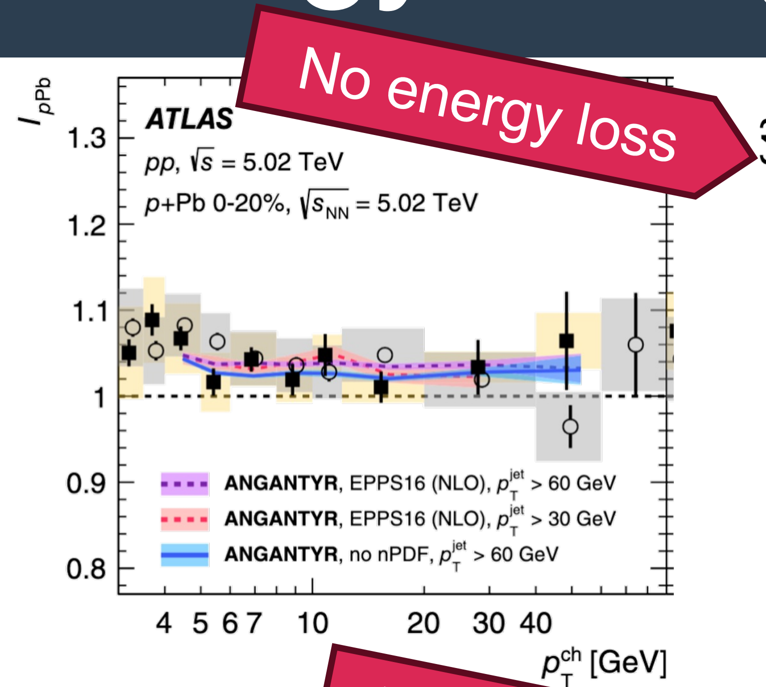

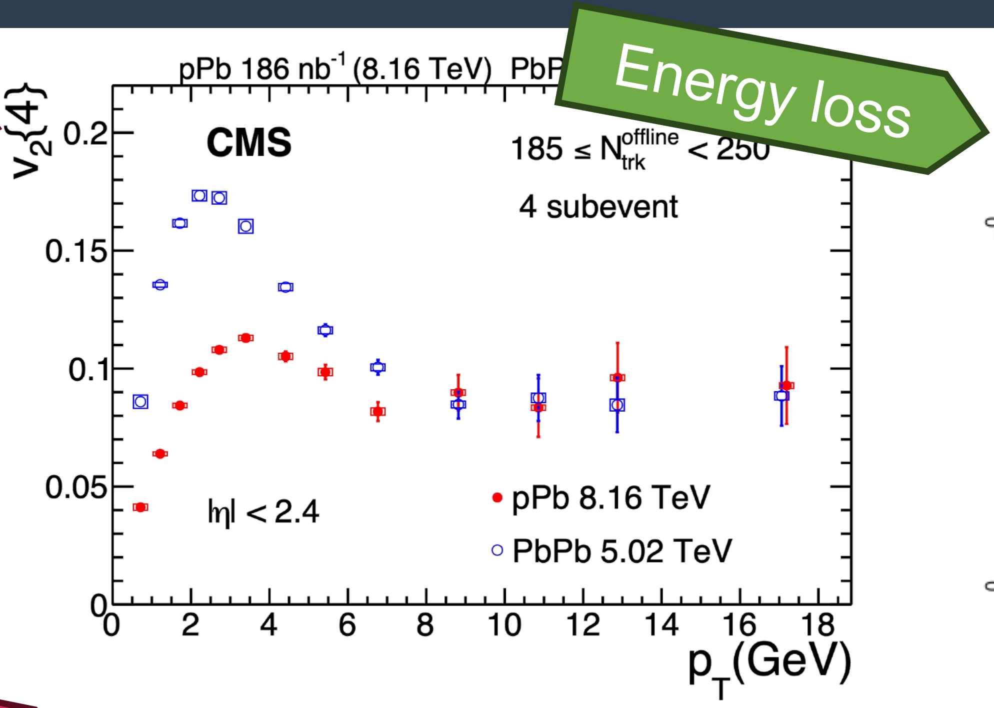

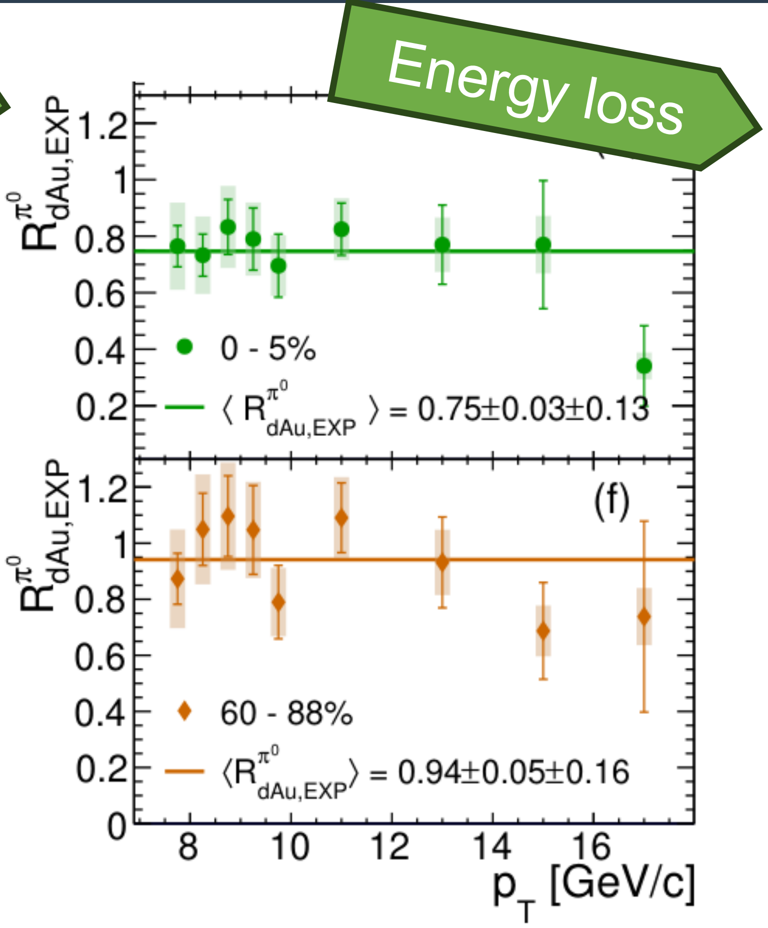

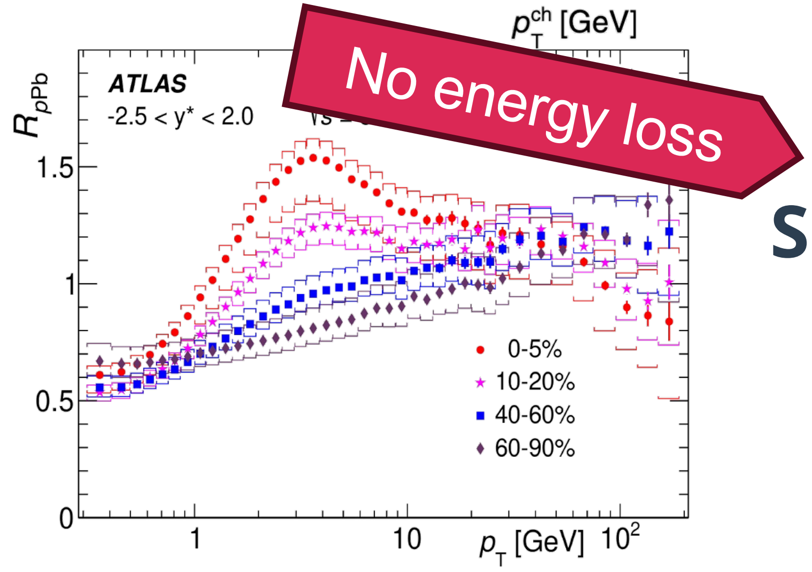

Small system suppression pattern is not clear

→ Theory input needed

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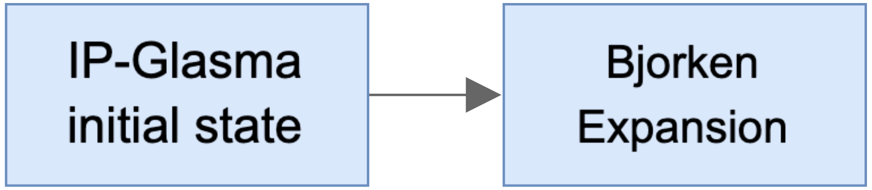
Physics model



CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

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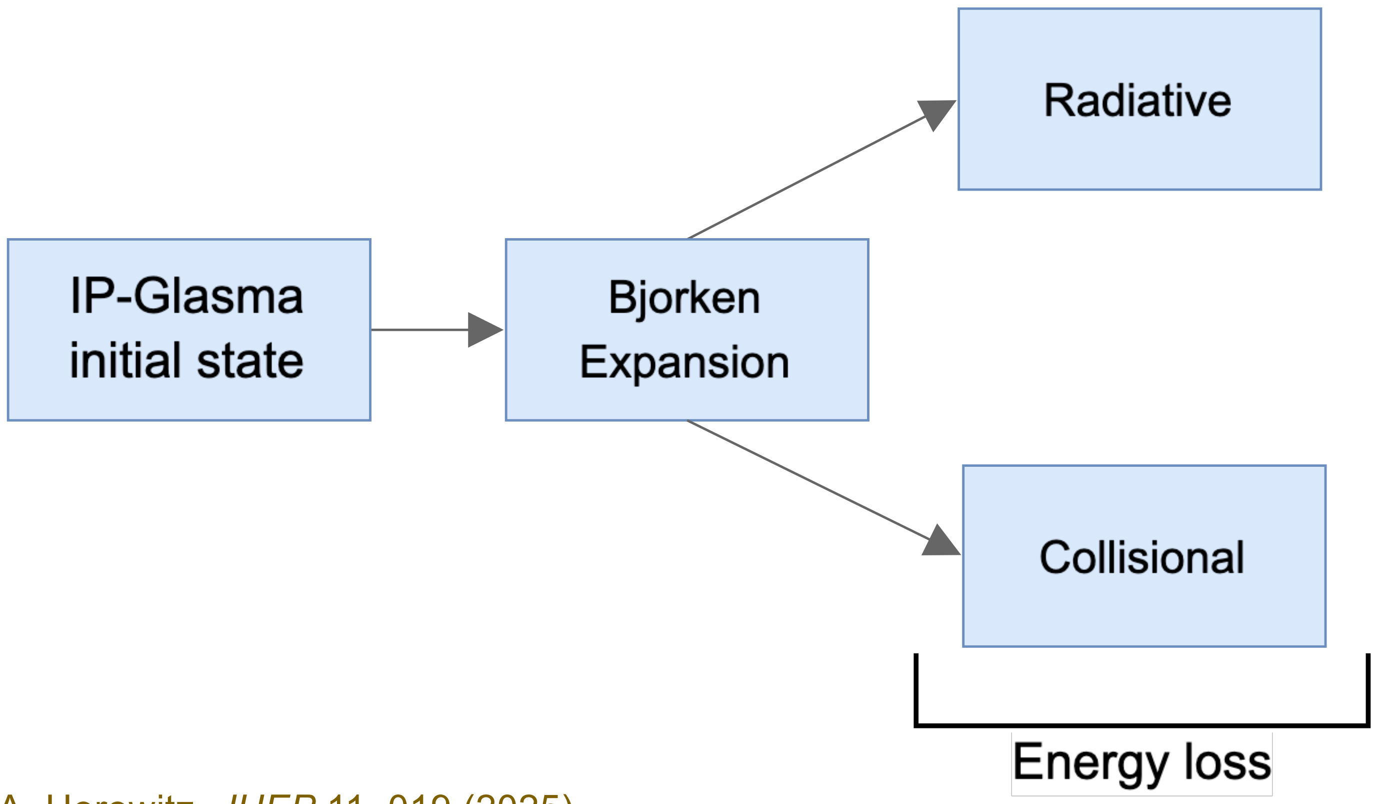

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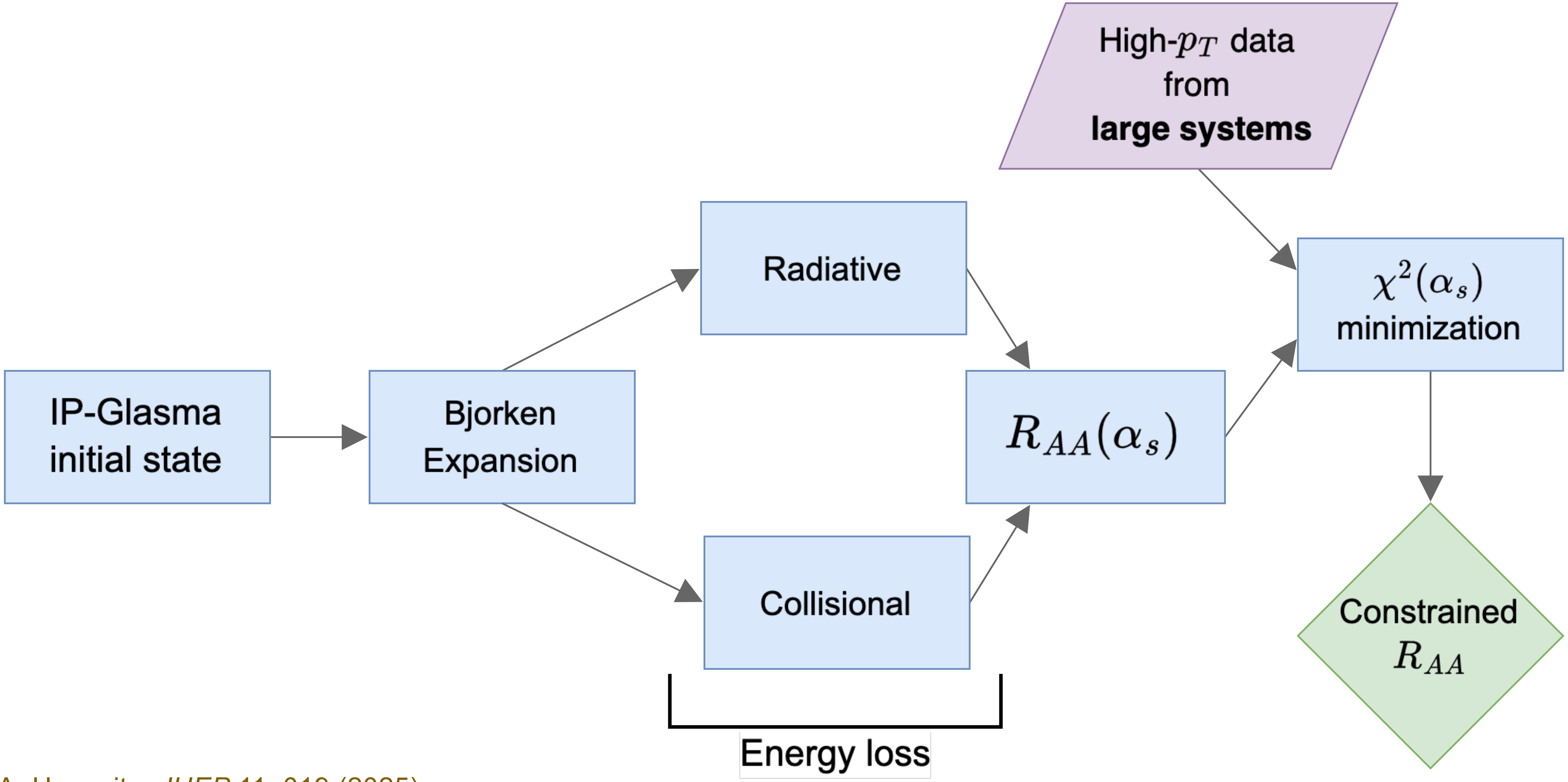

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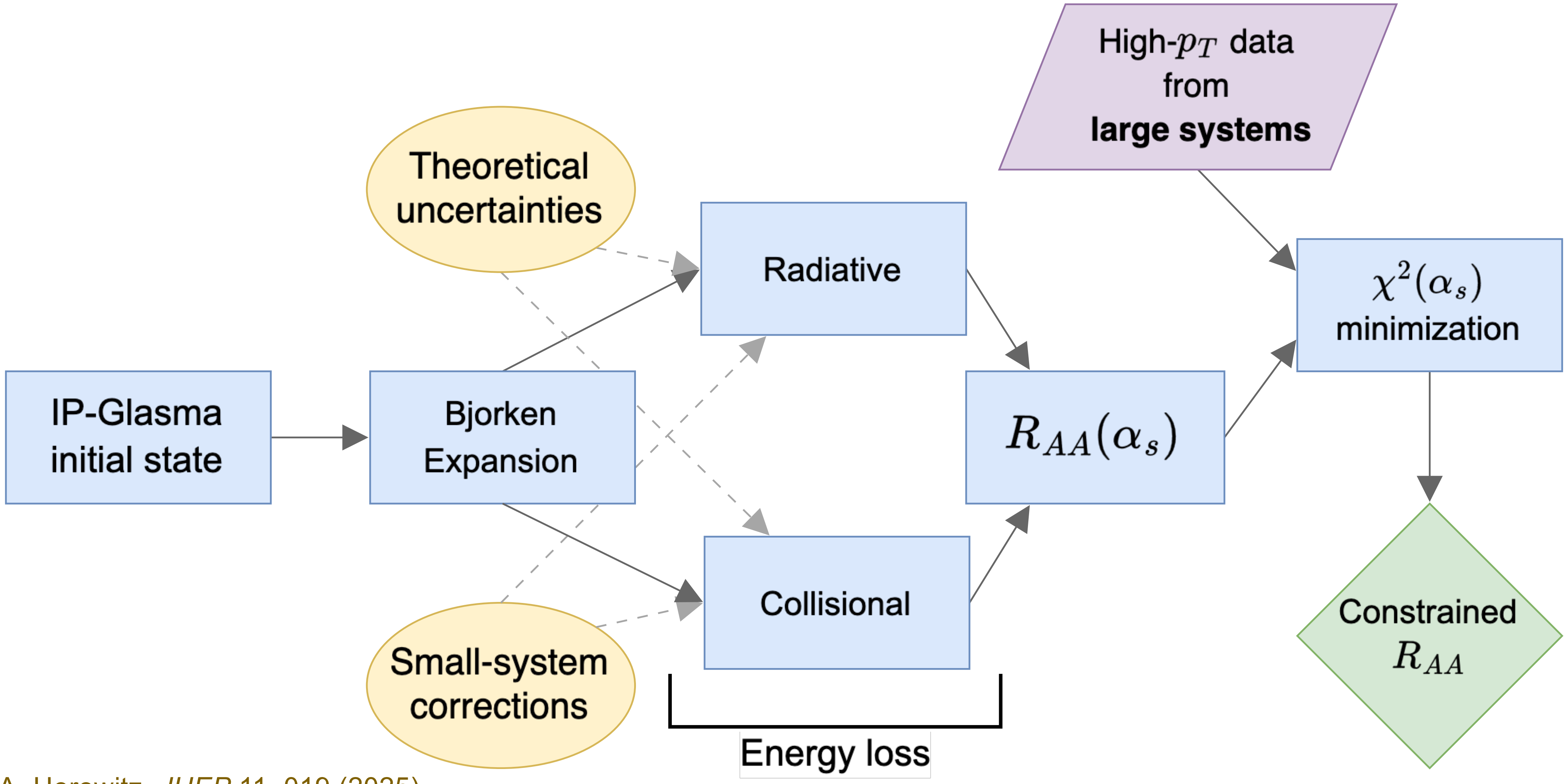

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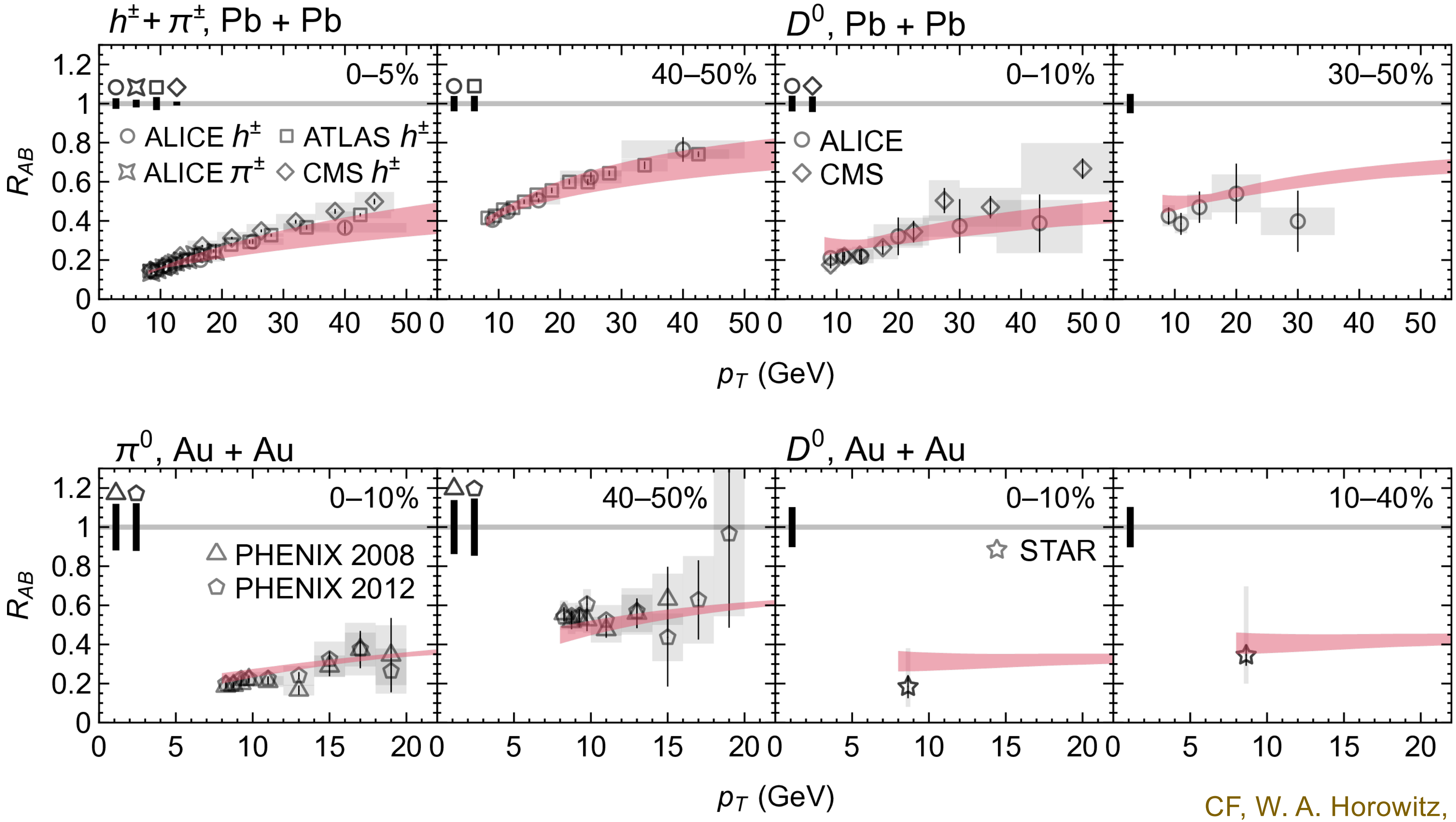
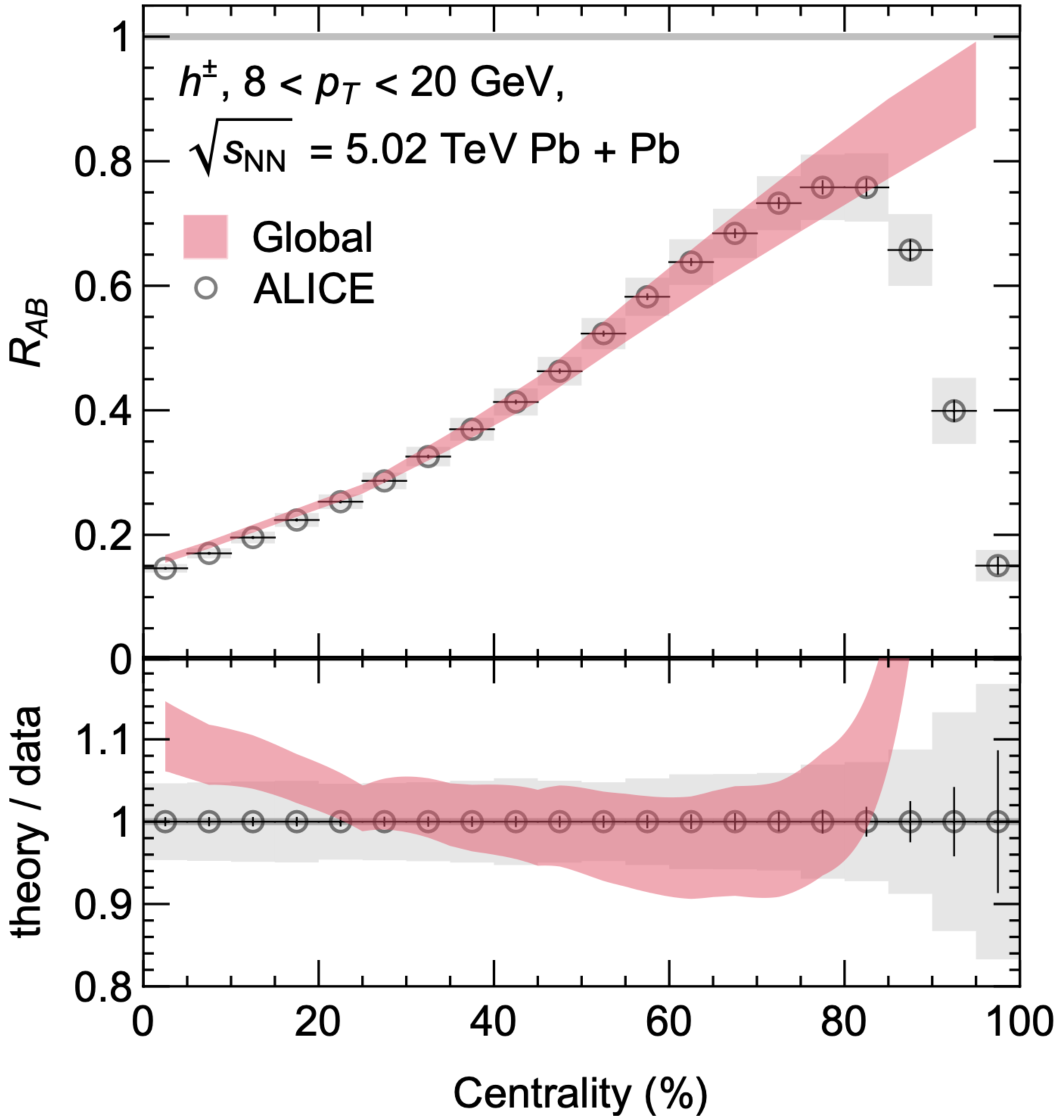
Sample of results post-fitting



Good agreement with all available single-inclusive high-p_T data

CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

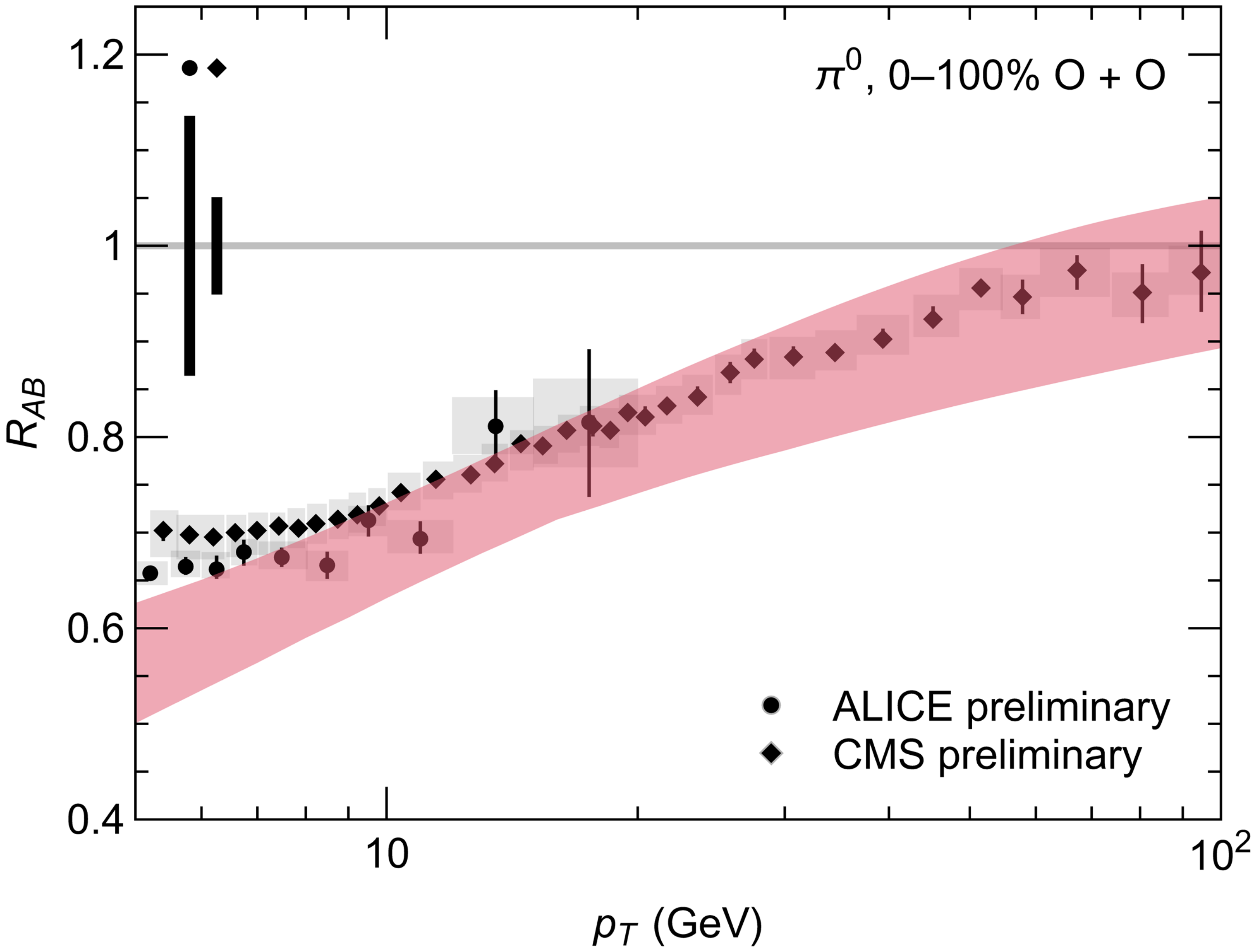
Centrality dependence



- Very good description of centrality dependence
- Above 80%, selection biases lead to large anomalous suppression
- Gives us confidence to extrapolate to small systems

CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

Predictions for light ions

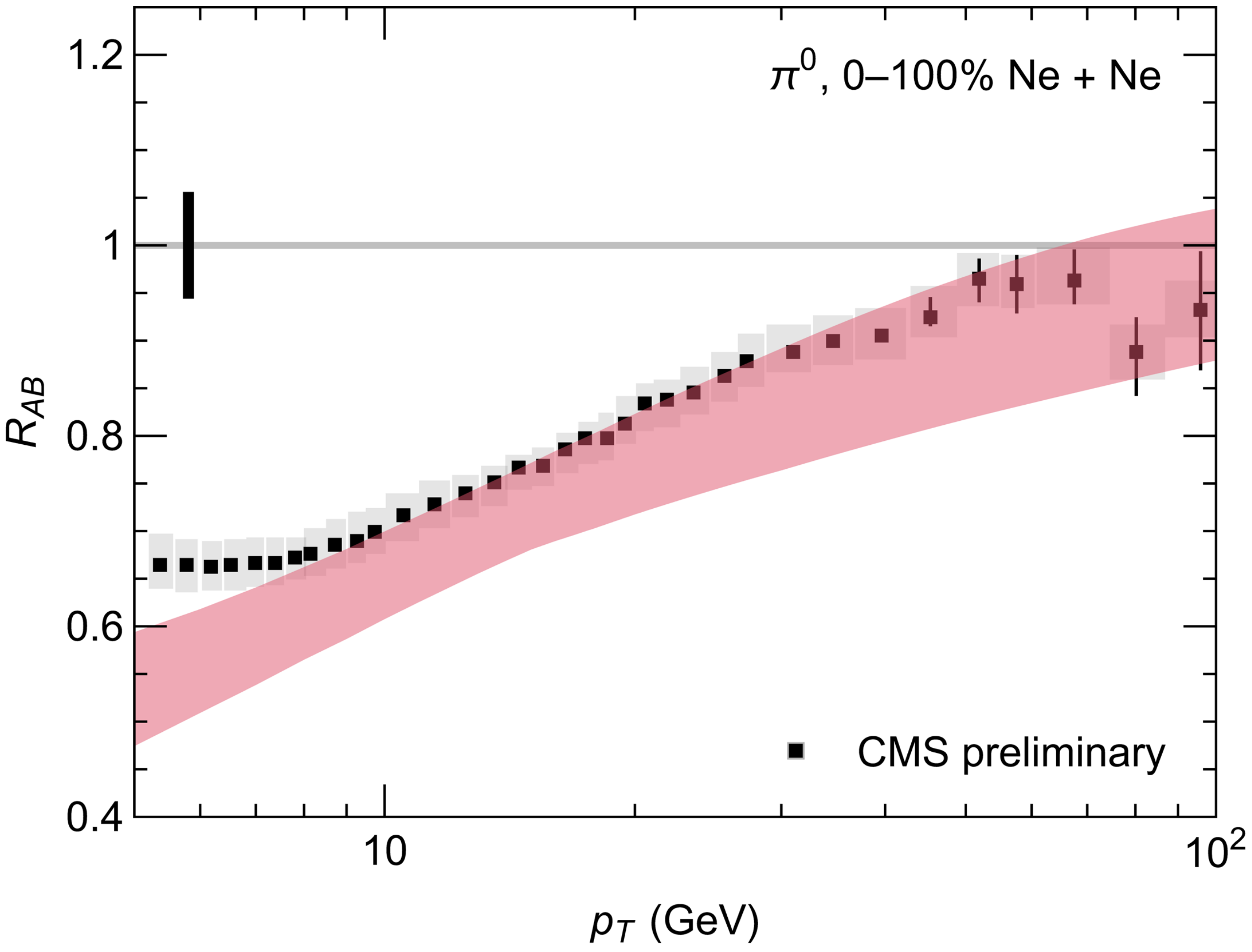


- Overall, good agreement between predictions and data!
- Slightly over-suppressed compared to data, but within theoretical uncertainties
- Missing physics for $p_T \lesssim 10$ GeV
  - Medium-modified hadronization (e.g. coalescence?)

CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

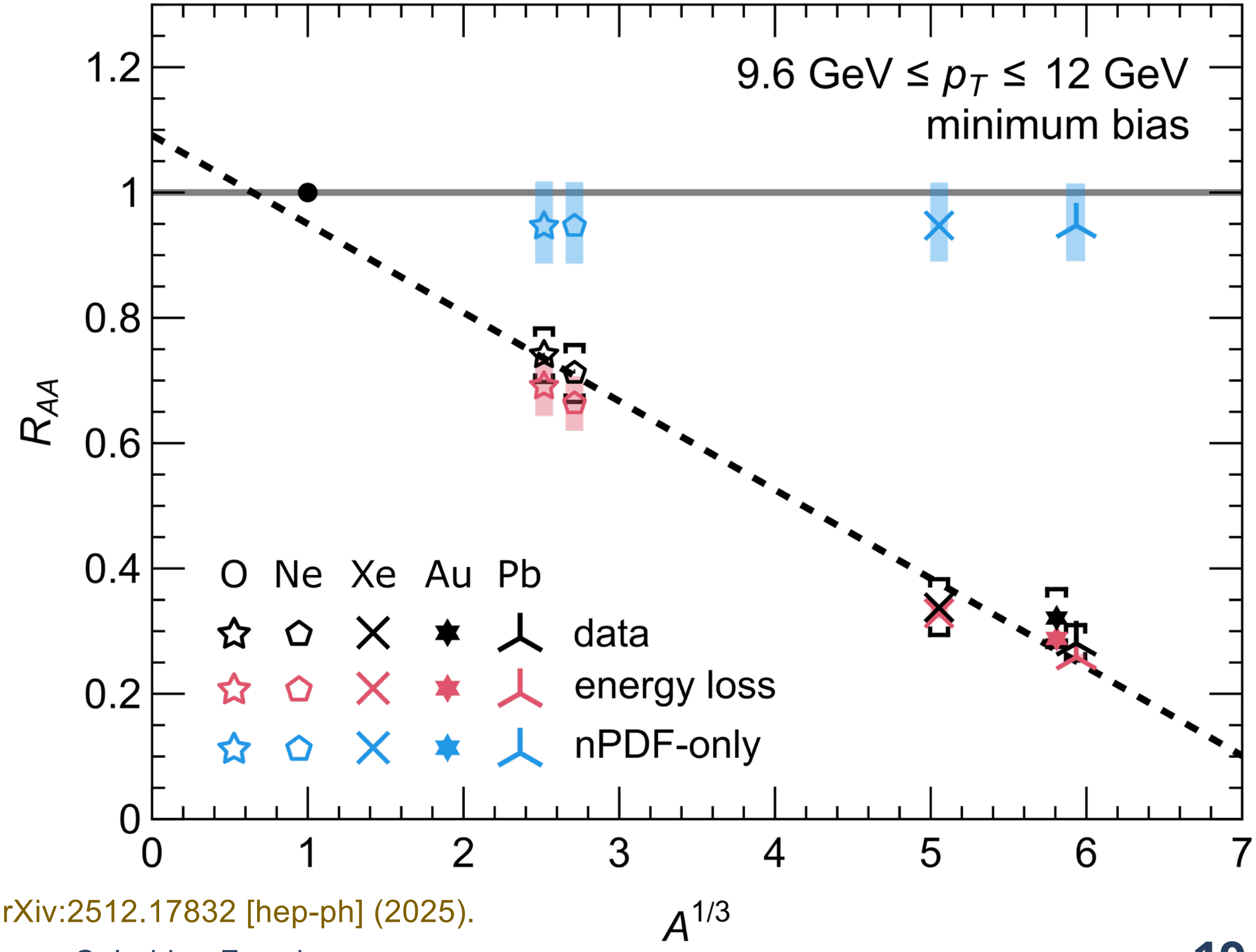
Predictions for light ions

See similar (small) increase in suppression in NeNe as data



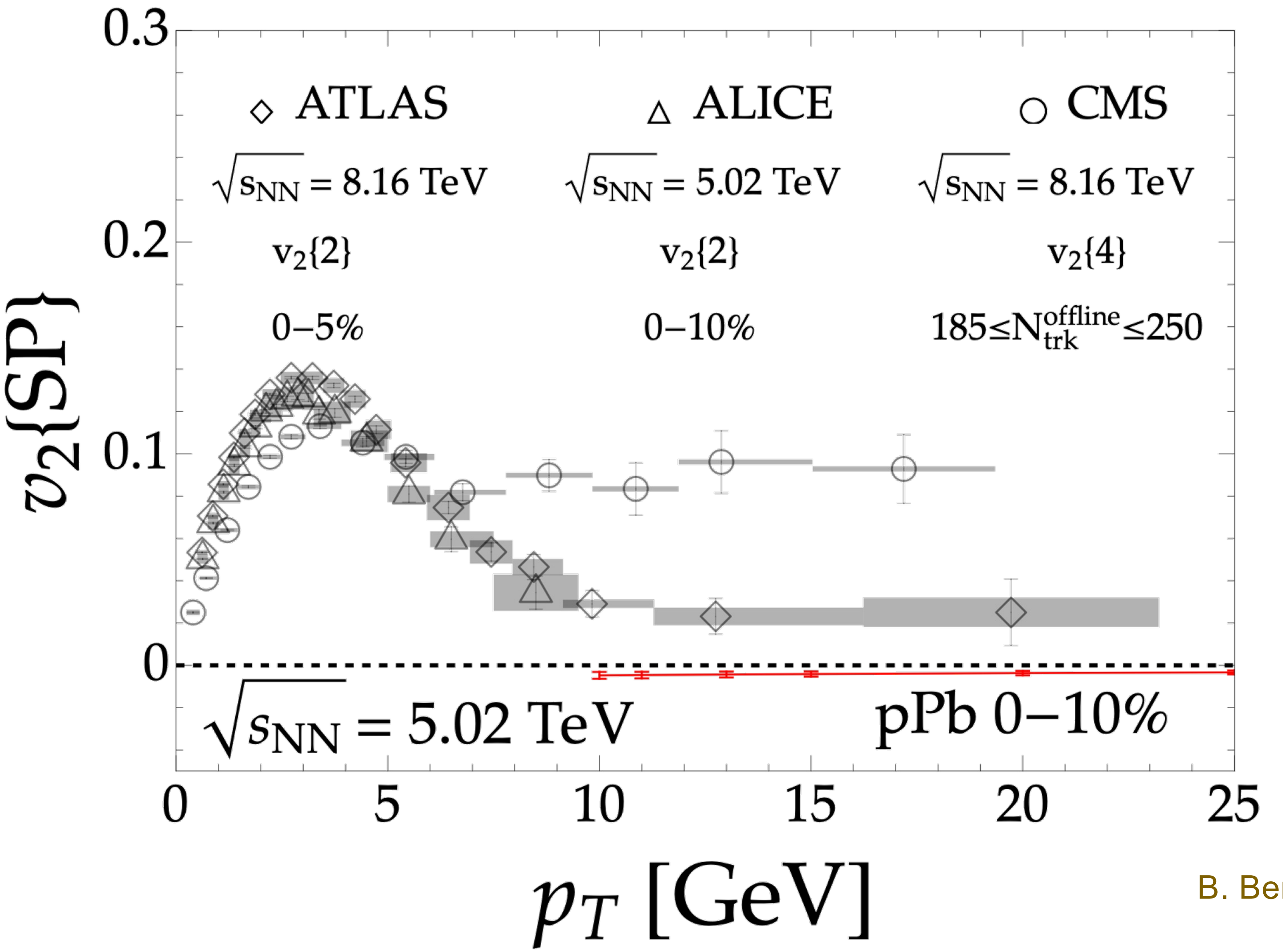
R_{AA} across system size

- Good agreement from our model across a wide range of systems (A =208 to A=16)
- Approximate linear dependence of R_{AA} in $A^{1/3}$



CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, arXiv:2512.17832 [hep-ph] (2025).

What about v_2?

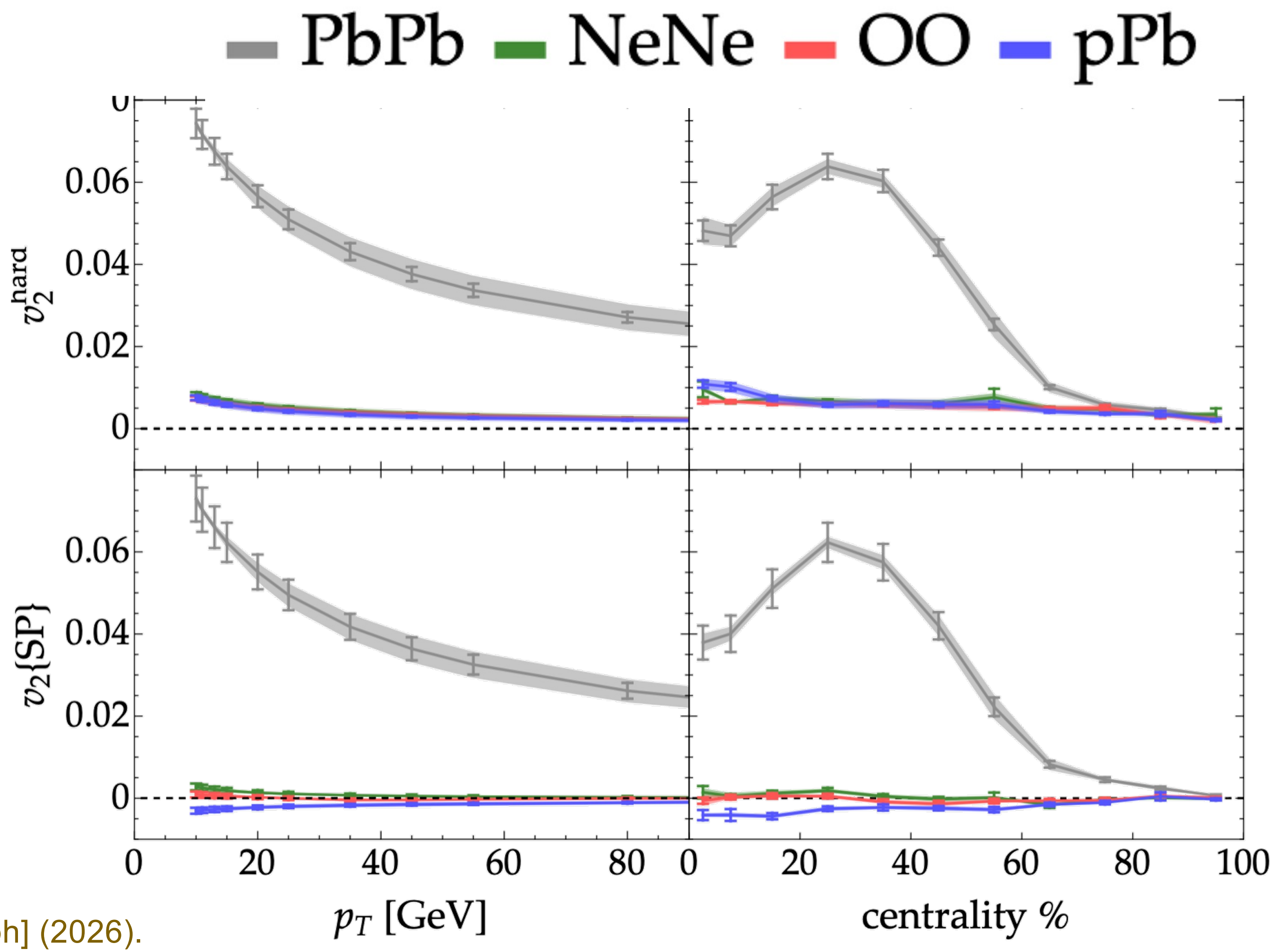


v_2 in p+Pb is confusing, both experimentally and theoretically

B. Bert, CF, and W. A. Horowitz, arXiv:2603.21861 [hep-ph] (2026).

v_2 in light ions?

- Small theory v_2 in all small systems
- Close to ***zero*** $v_2\{SP\}$ (what experimentalists measure)
- Measurement of high-p_T v_2 in OO would tell use whether the large v_2 in *p*Pb is due to *p*Pb specific complications or more generic



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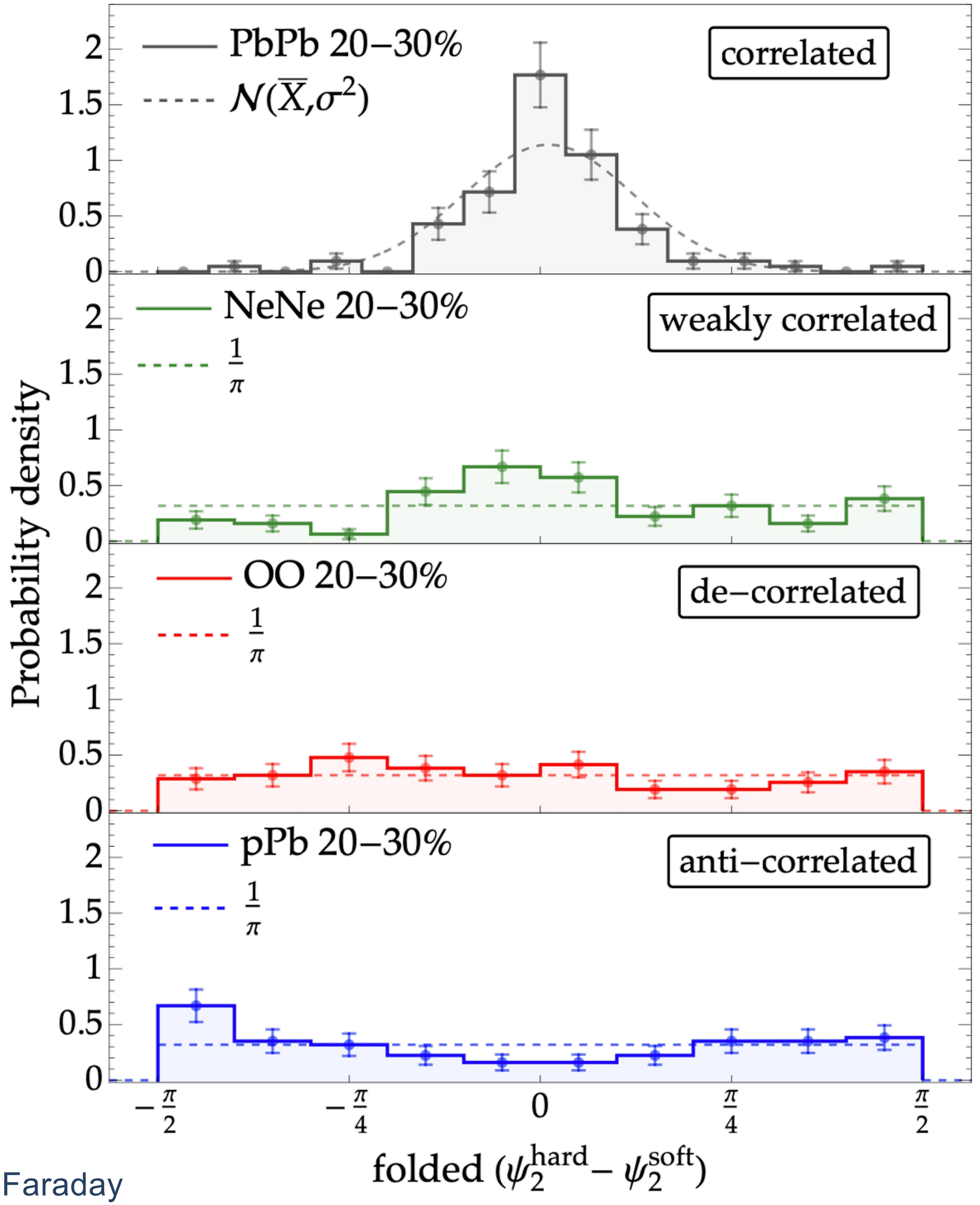
Why is $v_2 \approx 0$?

- ψ_2^{hard} and ψ_2^{soft} correlated in Pb + Pb
  - Uncorrelated or even anti-correlated in small systems (!)



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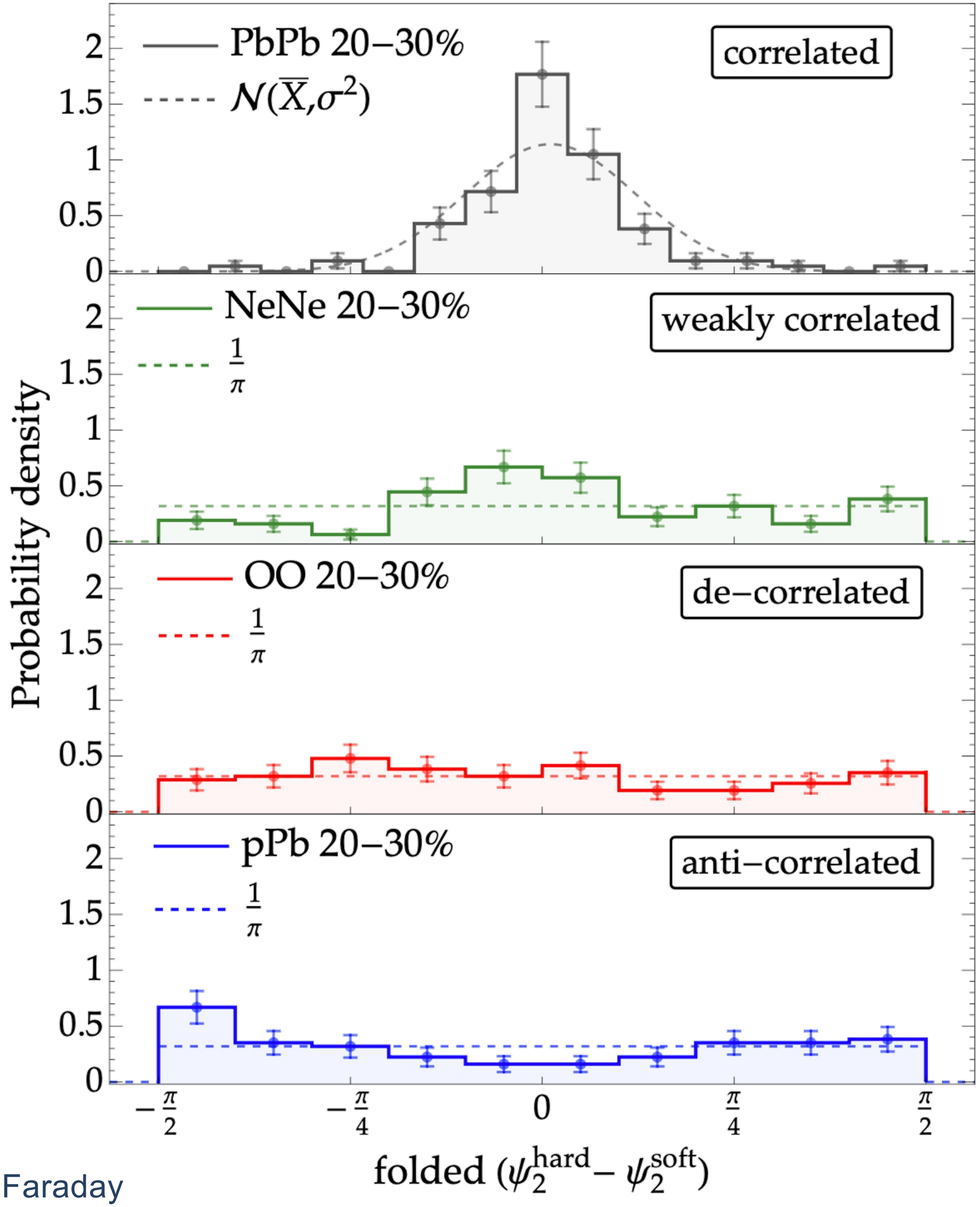

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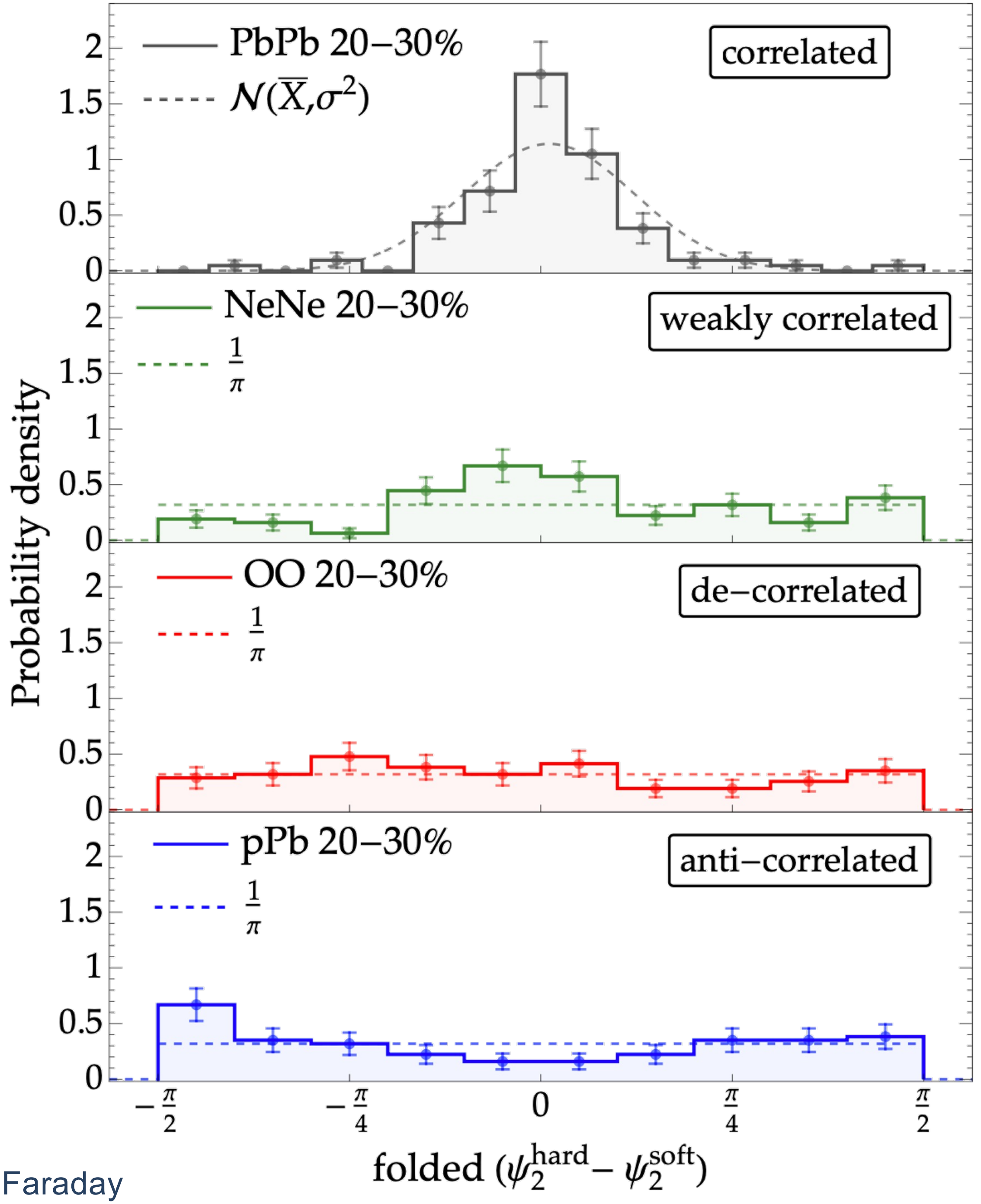
  - Uncorrelated or even anti-correlated in small systems (!)
- Not possible to create $v_2\{2\}$ in small systems from energy loss
- Likely culprit: initial state effects / hard-soft correlations
- Future: electroweak $v_2\{2\}$ measurement and modeling of hard-soft correlations



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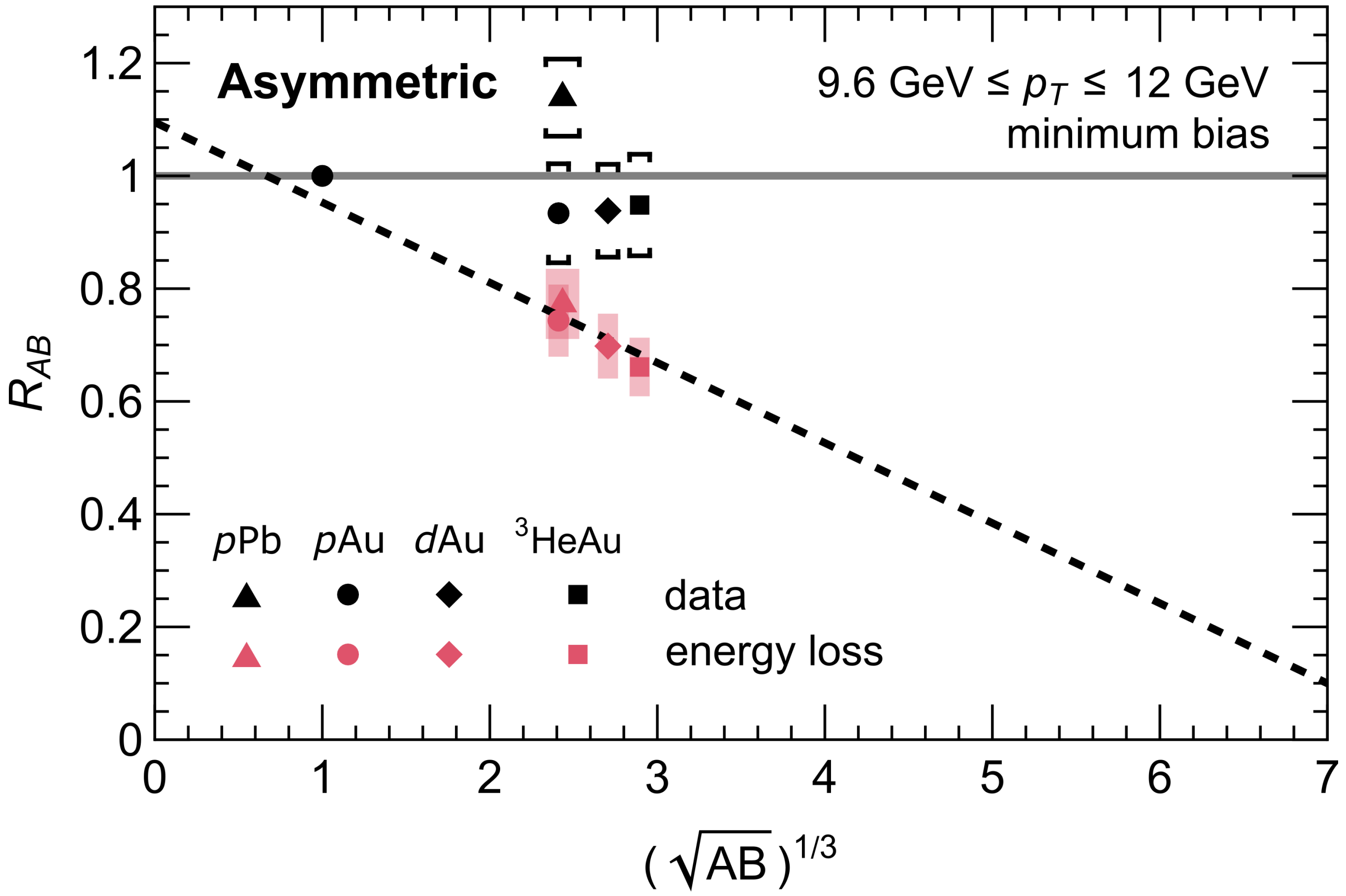
Can we go smaller? *pA, dA, …*



- Compare to minimum bias available asymmetric small system data
- Model shows qualitatively different trend to experiment

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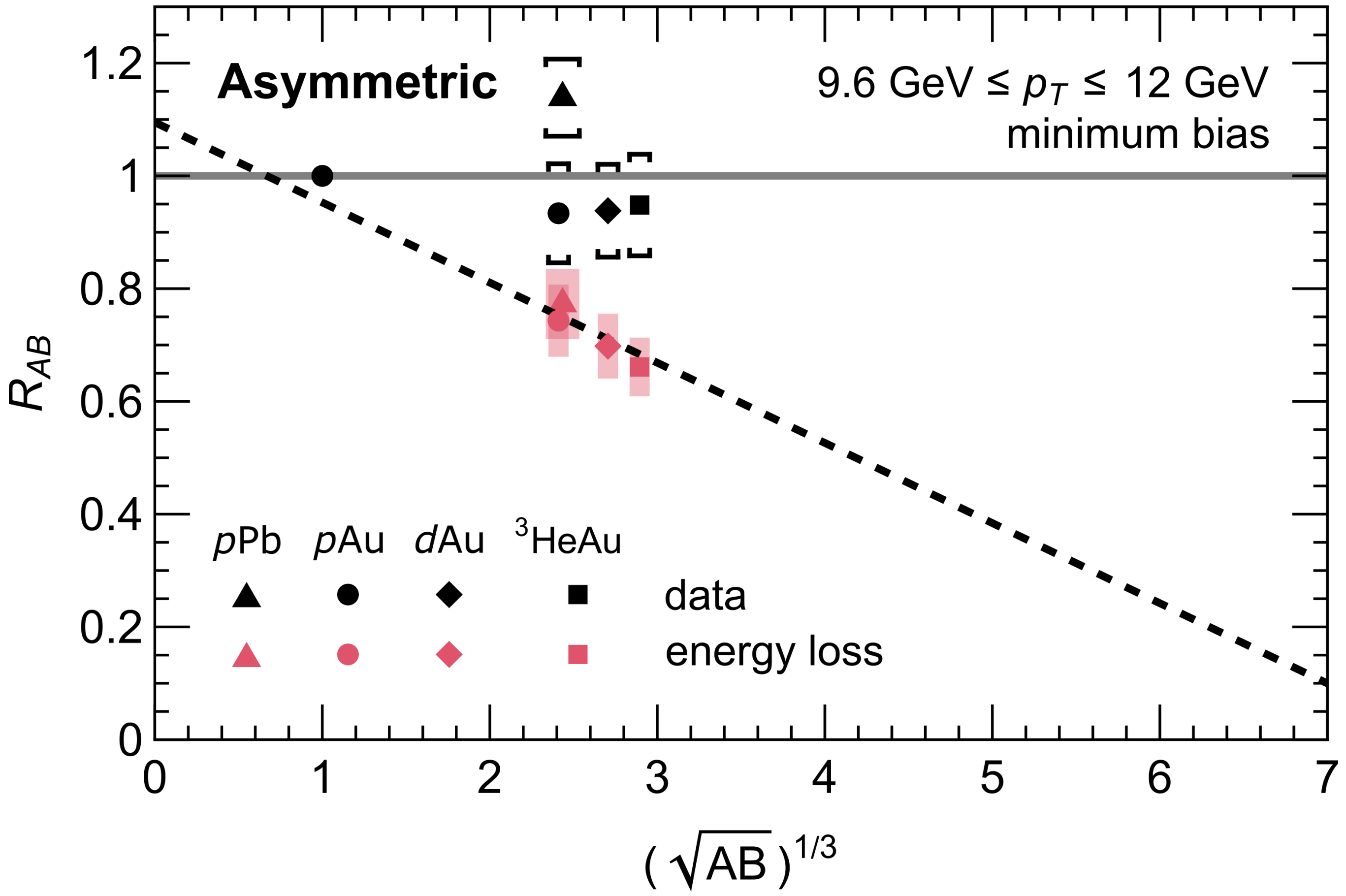
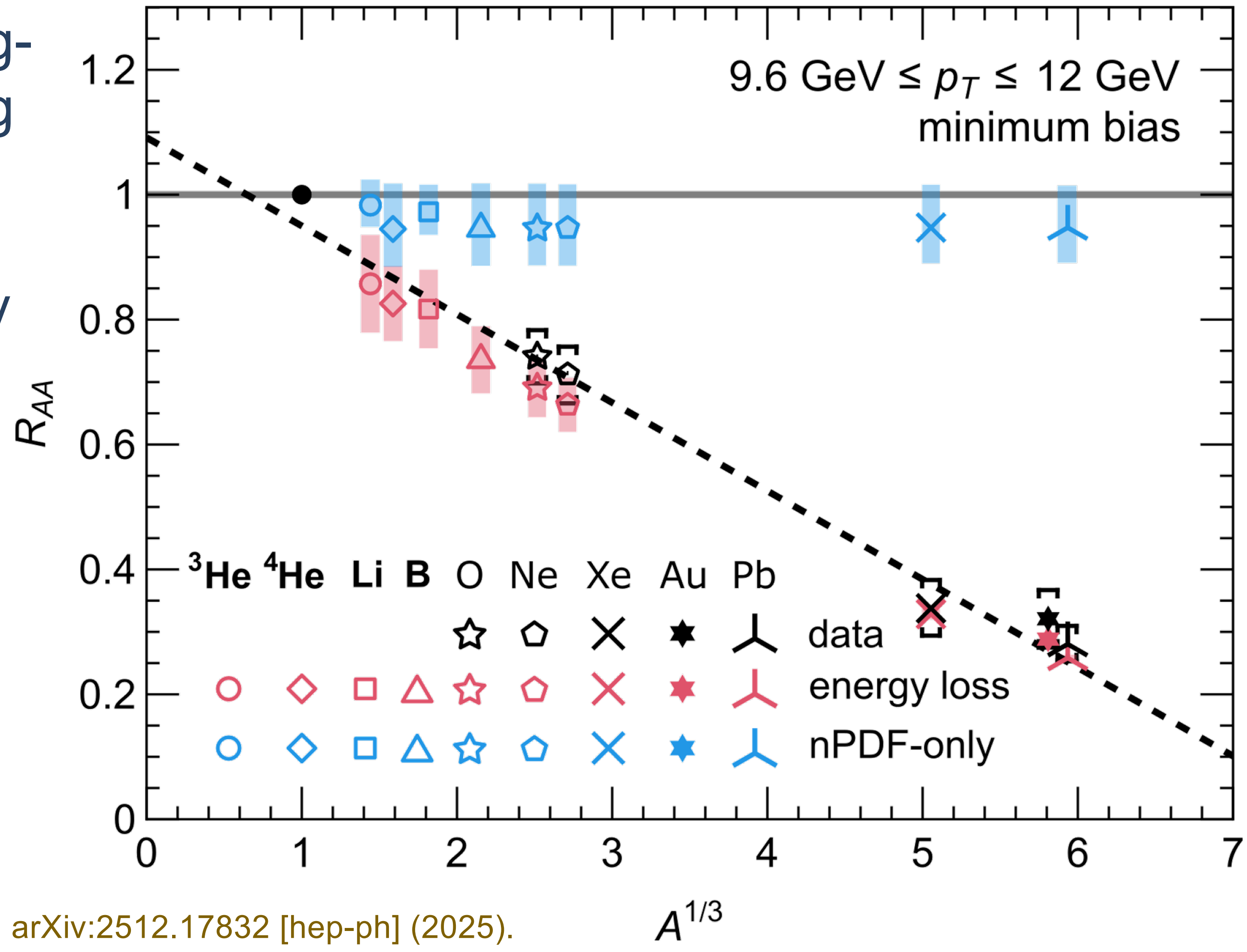


- Compare to minimum bias available asymmetric small system data
- Model shows qualitatively different trend to experiment
- Asymmetric systems have lots of complications:
  - Breaking of boost invariance
  - Single proton responsible for both hard and soft production (sensitive to hard-soft correlations and proton structure)
  - Large CNM effects

CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, arXiv:2512.17832 [hep-ph] (2025).

Can we go smaller? – Even lighter ions

- ^{3}He and ^{6}Li offer the most "bang-for-buck" in terms of discovering energy loss in very small systems
- ^{6}Li and ^{10}B offer ability to clearly differentiate between energy loss and initial state effects



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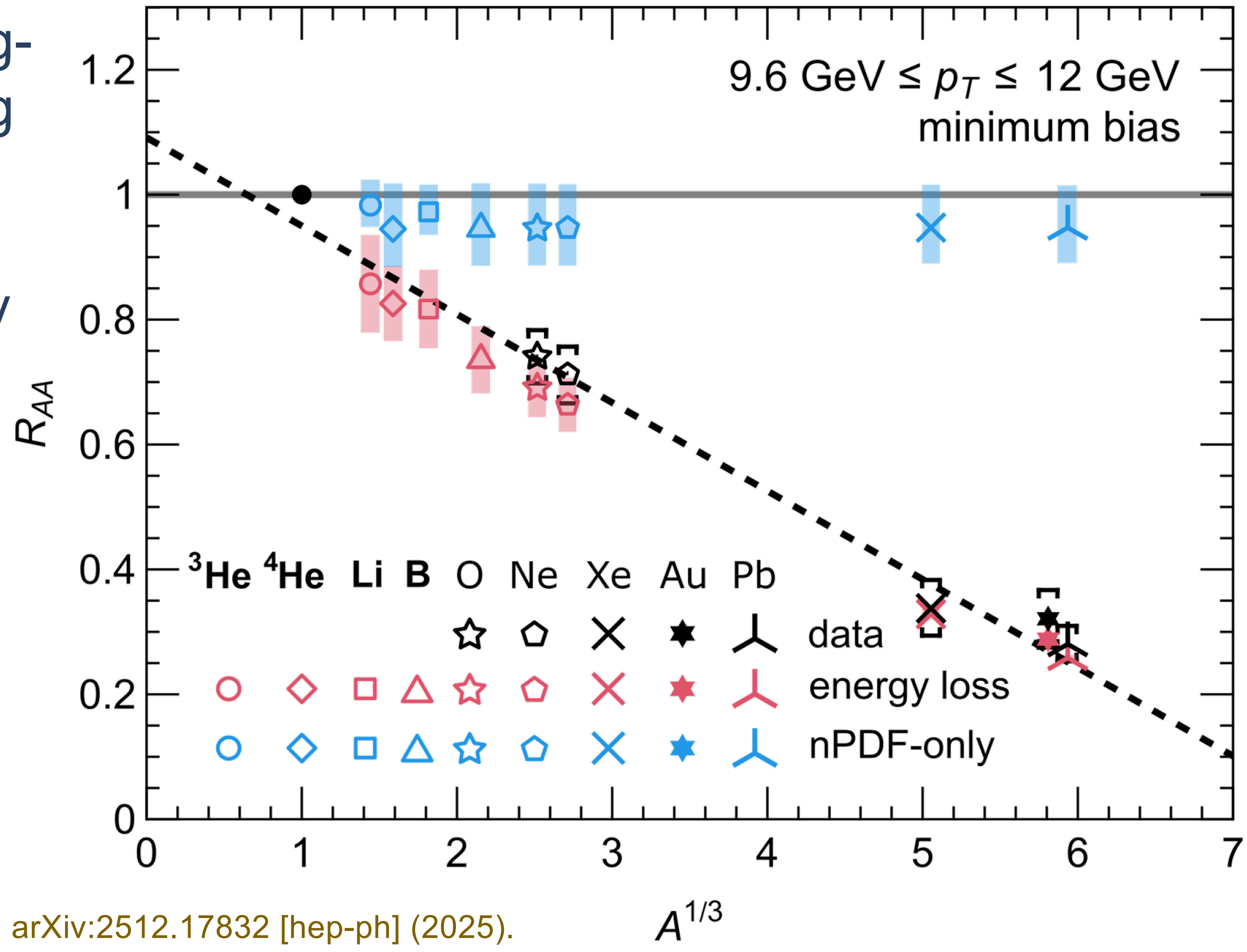
- ^{6}Li and ^{10}B offer ability to clearly differentiate between energy loss and initial state effects
- Such systems could be considered for future LHC runs, similar to Ne+Ne; $\lesssim$ 1 day of runtime would yield enough statistics



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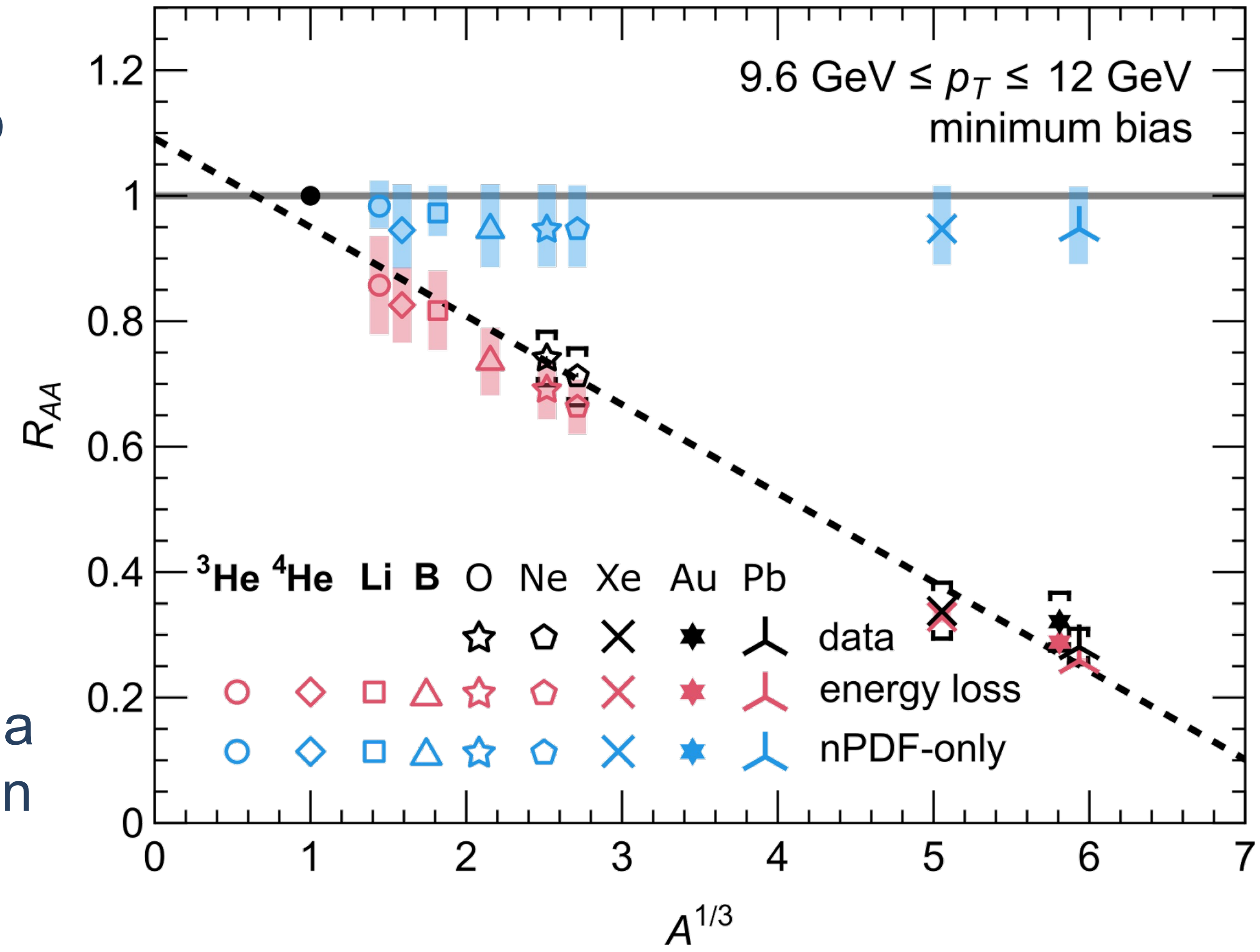
Conclusions and outlook

- Energy loss in small systems is the **key** to understanding whether QGP forms.
- OO & NeNe form a crucial stepping stone towards even smaller systems

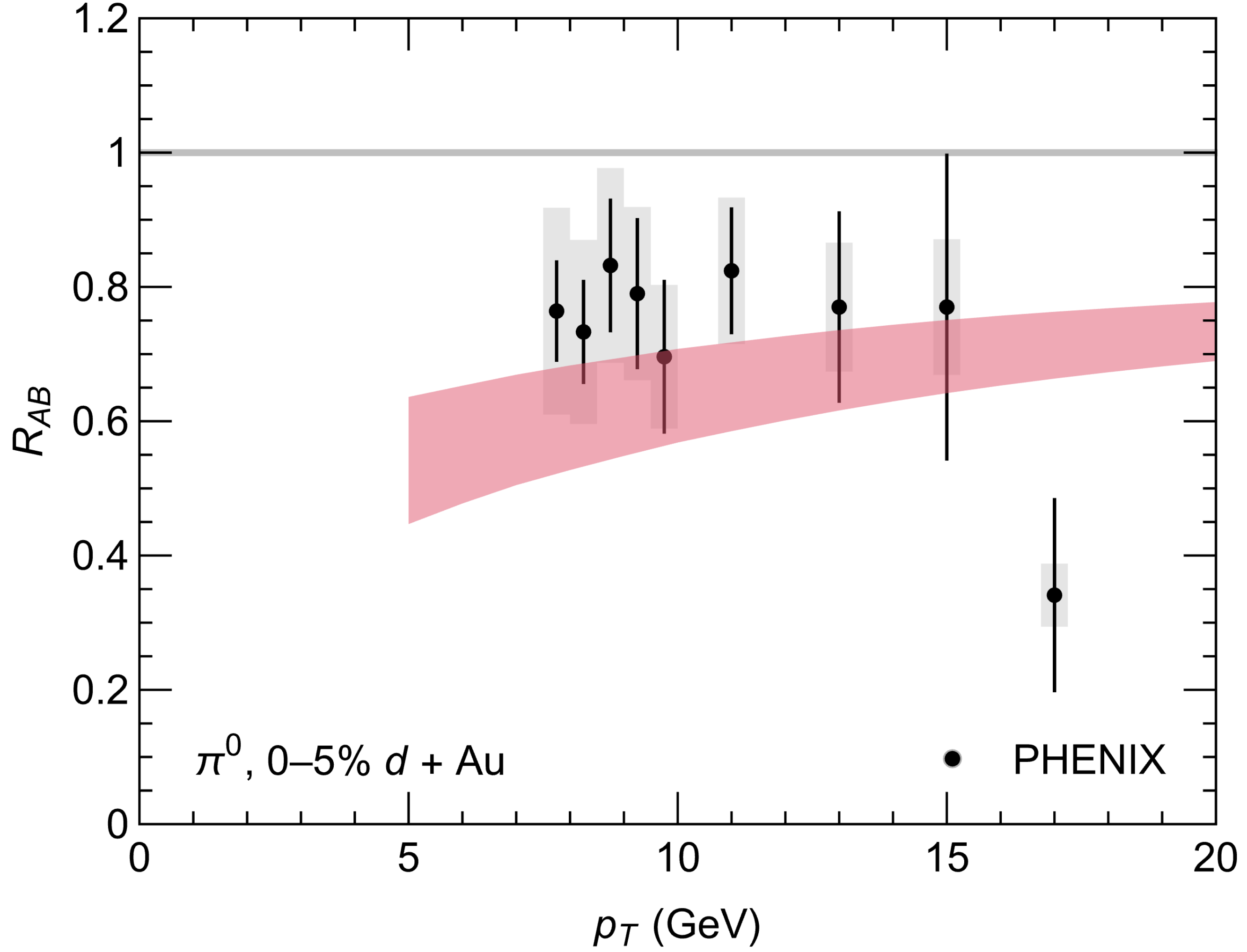
- Evidence that high-p_T v_2 is not a clear signature of partonic energy loss in small systems

- *p*Pb is very complicated; lighter-ions offer a unique opportunity to observe collectivity in systems with only ~ 100 particles



Backup

Can we go smaller? *d*Au



- Good agreement with *d*Au double ratio

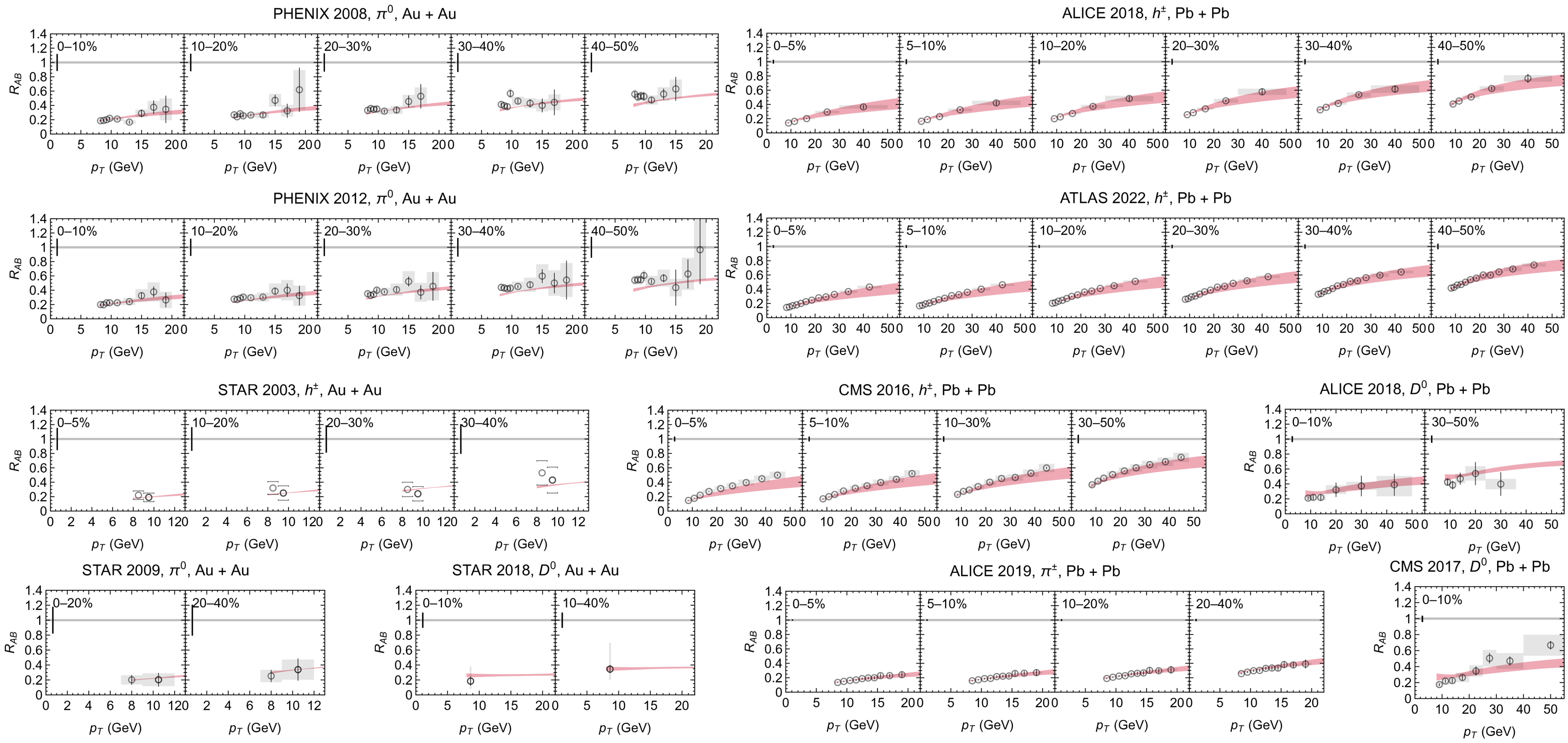
$$R^{\pi^0}_{d\mathrm{Au}}/R^{\gamma}_{d\mathrm{Au}}$$

- Reduced dependence on the Glauber model for normalization

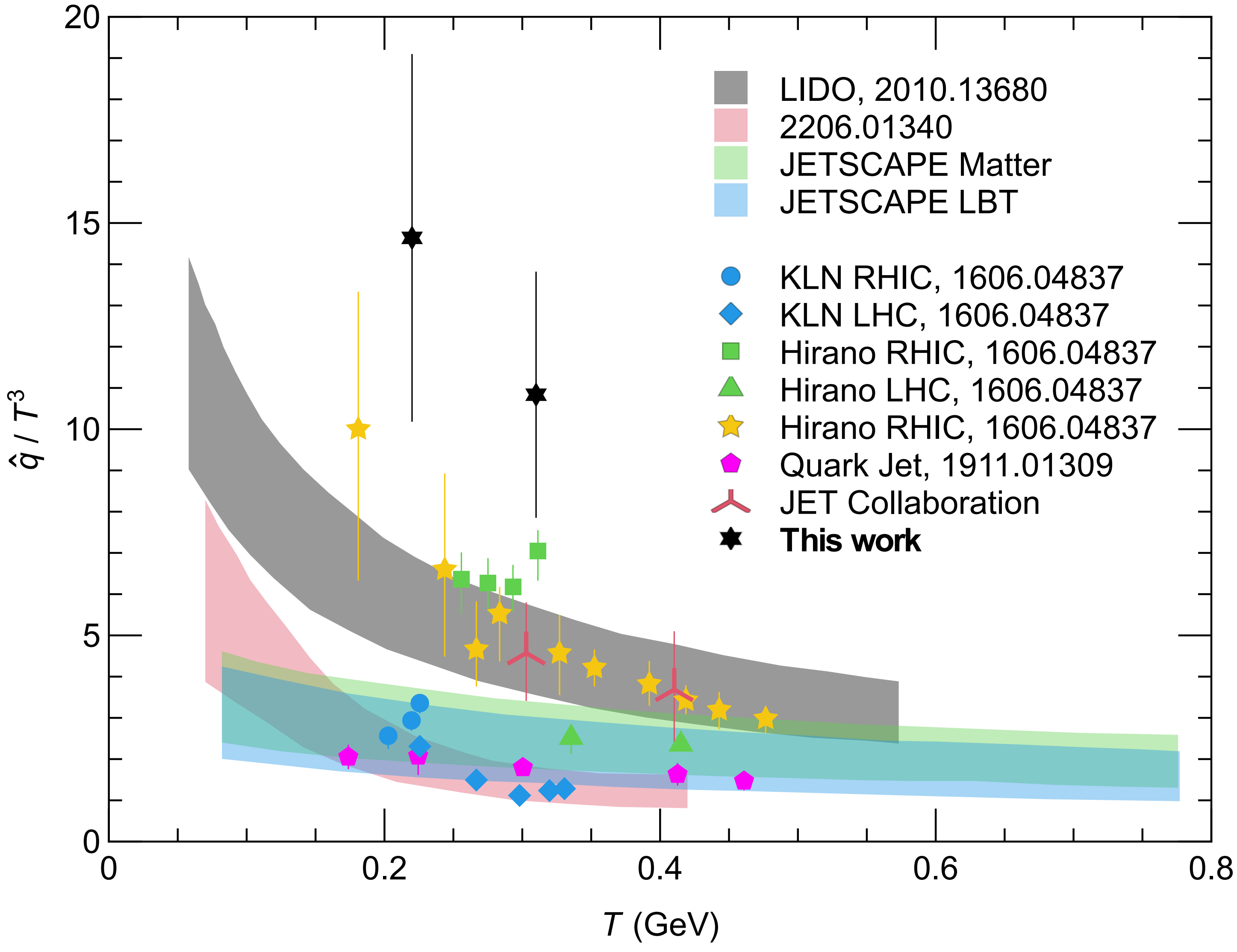
- Similar photon-normalized measurement in *p*Pb would be extremely interesting

CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

All results post-fitting, 295 data points

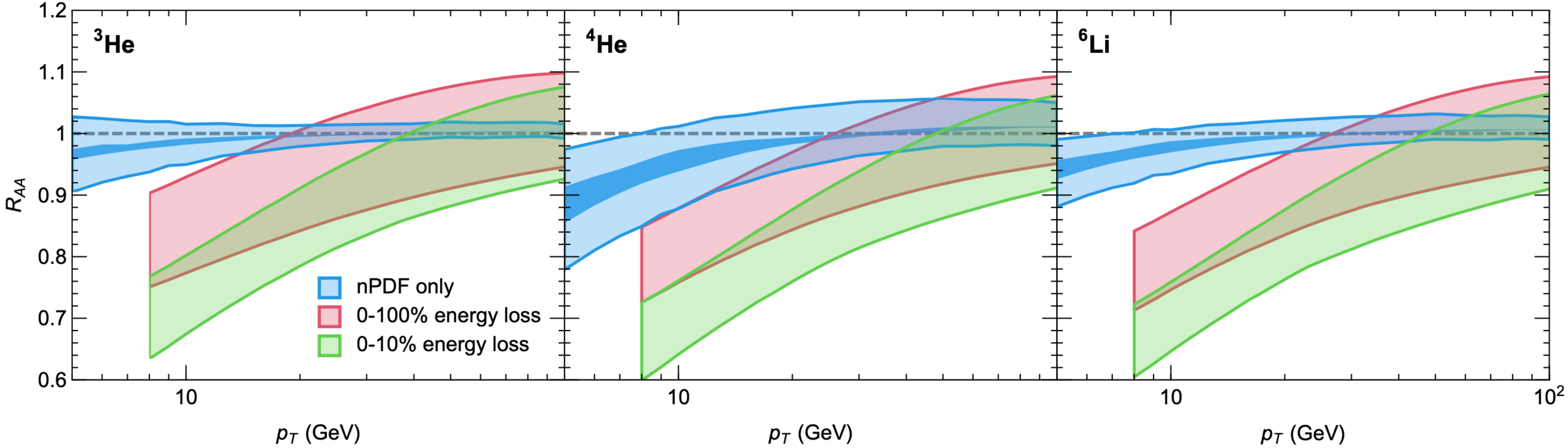


Extraction of $\hat{q}$



- **NB**: almost all uncertainty is due to theoretical uncertainties and **not** from the extraction procedure
- Wide range of extracted $\hat{q}$ from different models

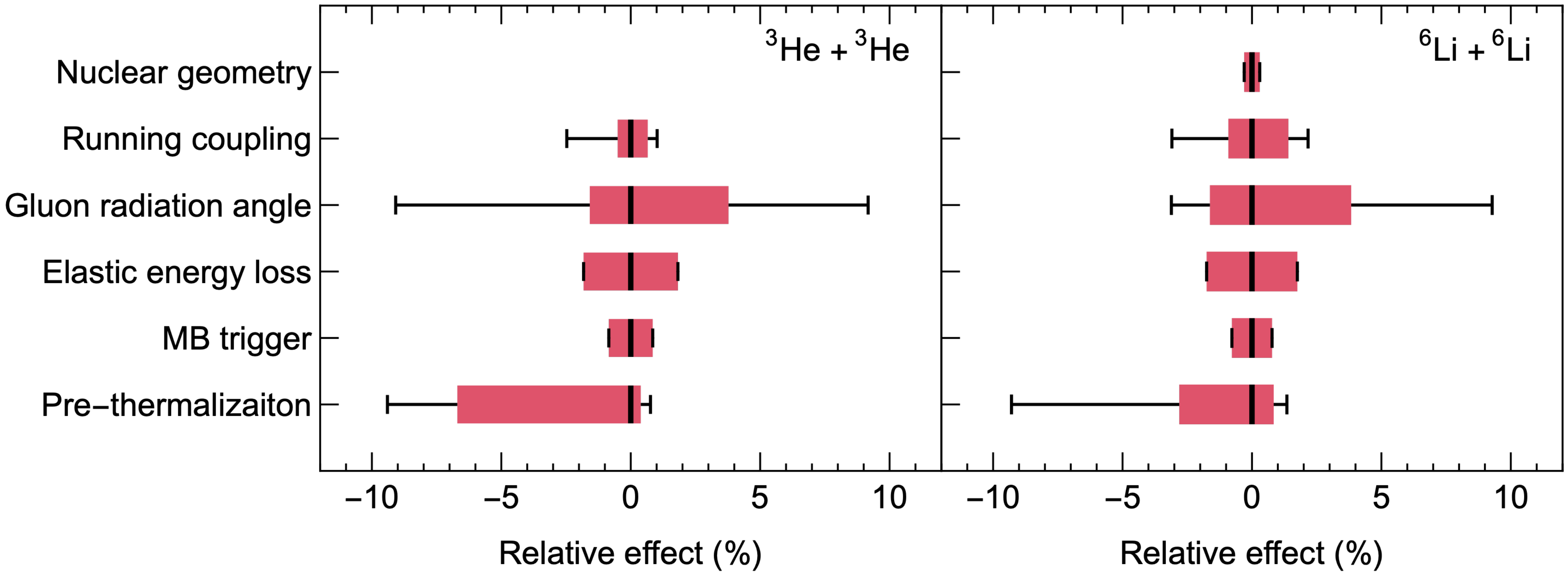
Centrality cut small systems?



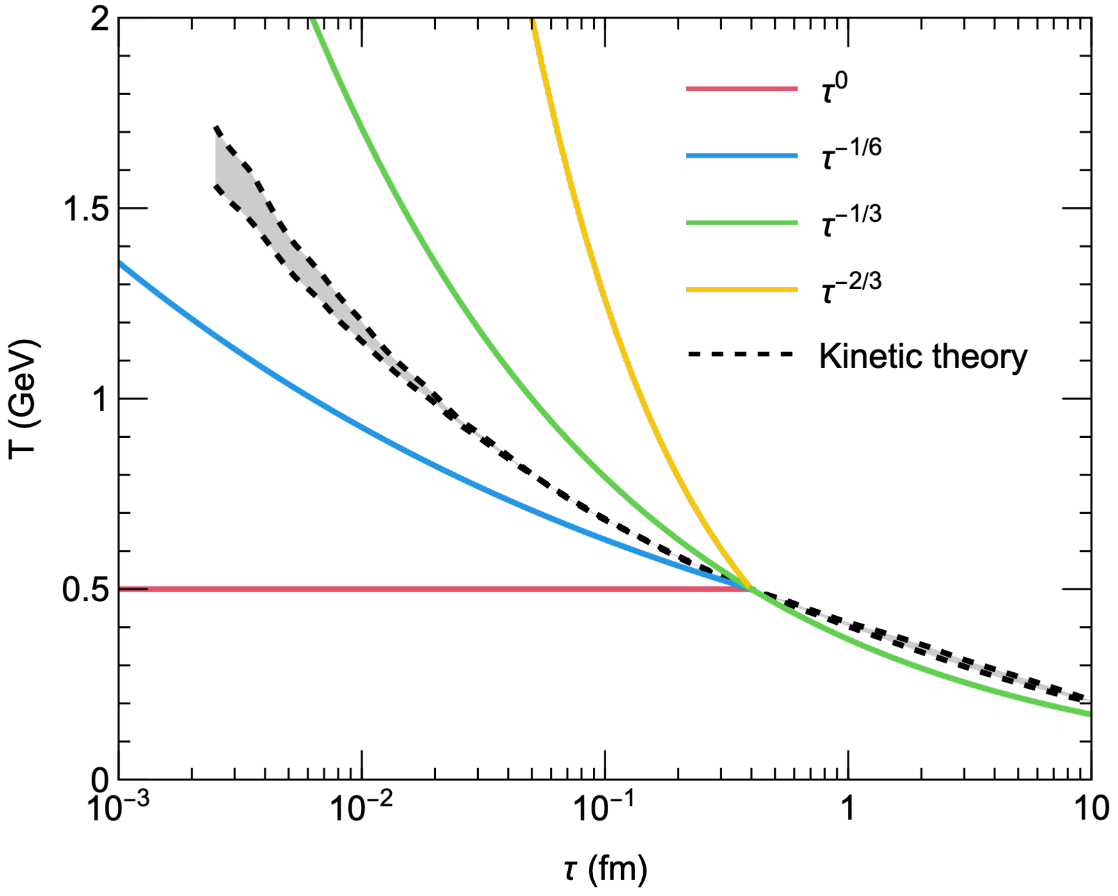
- Small window around ~8-20 GeV where energy loss should be measurable in min. bias
- If we can tackle centrality cuts in small systems, then any of these systems would have measurable energy loss

CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, arXiv:2512.17832 [hep-ph] (2025).

Summary of uncertainties

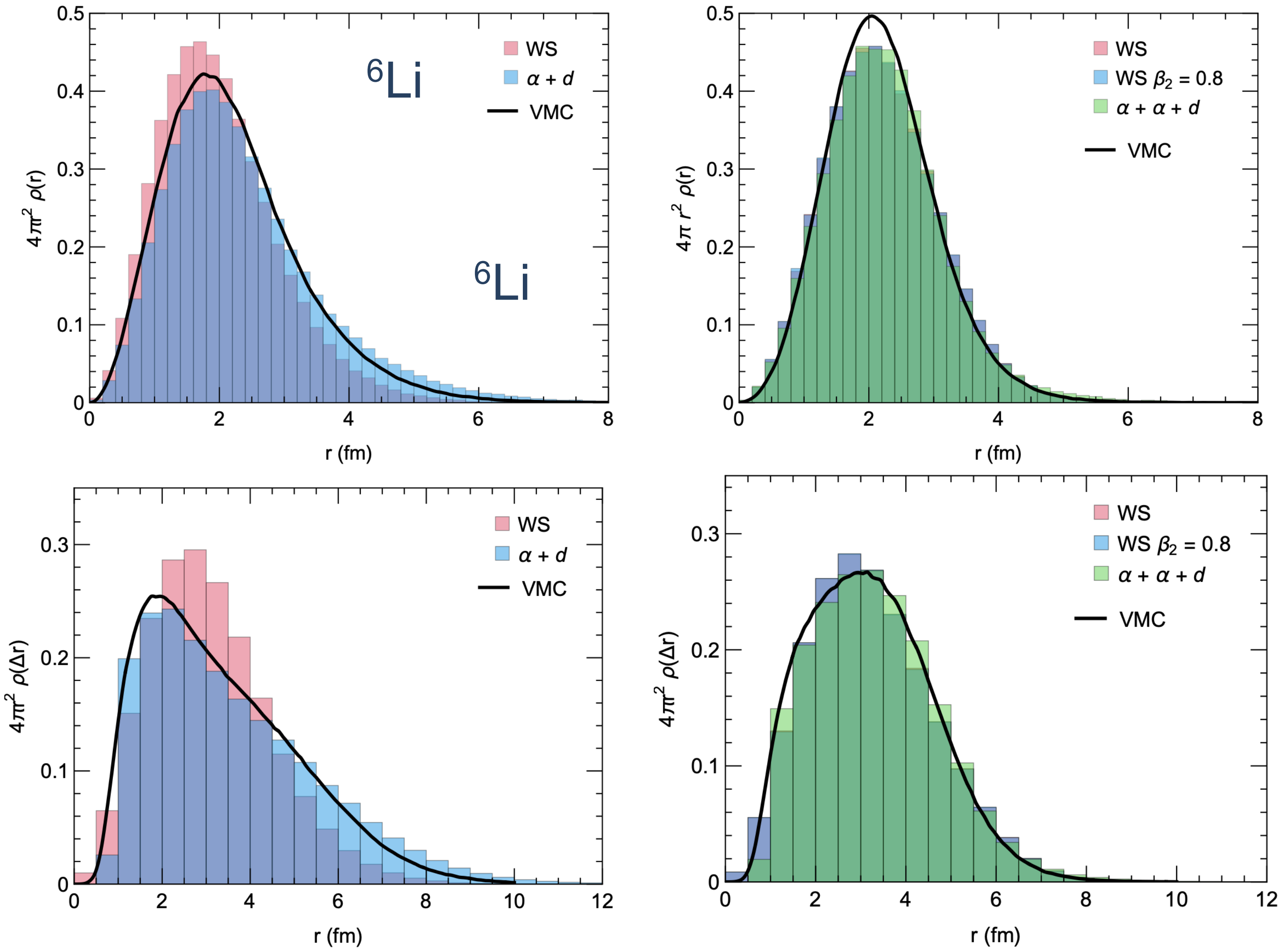


Prethermalization time “temperature”



- Consider the effect of the following different assumptions about temperature before thermalization as an uncertainty band on our predictions

Initial nuclear geometry



- Consider the uncertainty for the nuclear geometry in ^{6}Li and ^{10}B

Initial nuclear geometry

- Effect on the RAA is insensitive to nuclear geometry